

Genesys Aerosystems

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S-TEC 3100 Digital Flight Control System AEA Training Session

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Utilizing Genesys Software Tools

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TOPICS

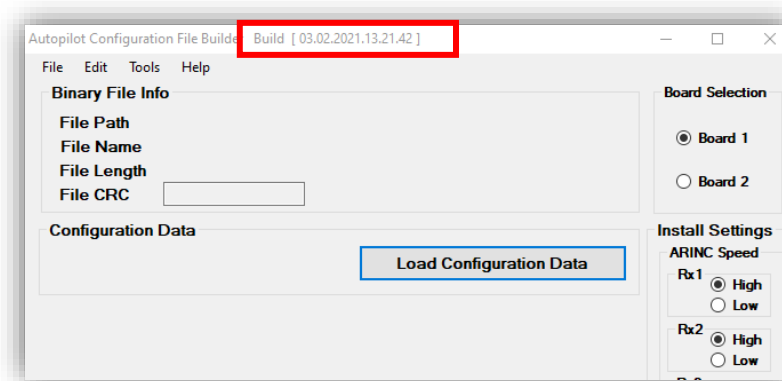
- Setting Up The Can Converter for AP Config Builder and TSO Vista
- AP Config Builder
- Troubleshooting **ATTITUDE FAIL** message
Stuck in **ADAHRS Initialize** or **Initialize**
- Troubleshooting **AP FAIL** message
Power On Self Test (POST) failures encountered during a 3100 DFCS installation



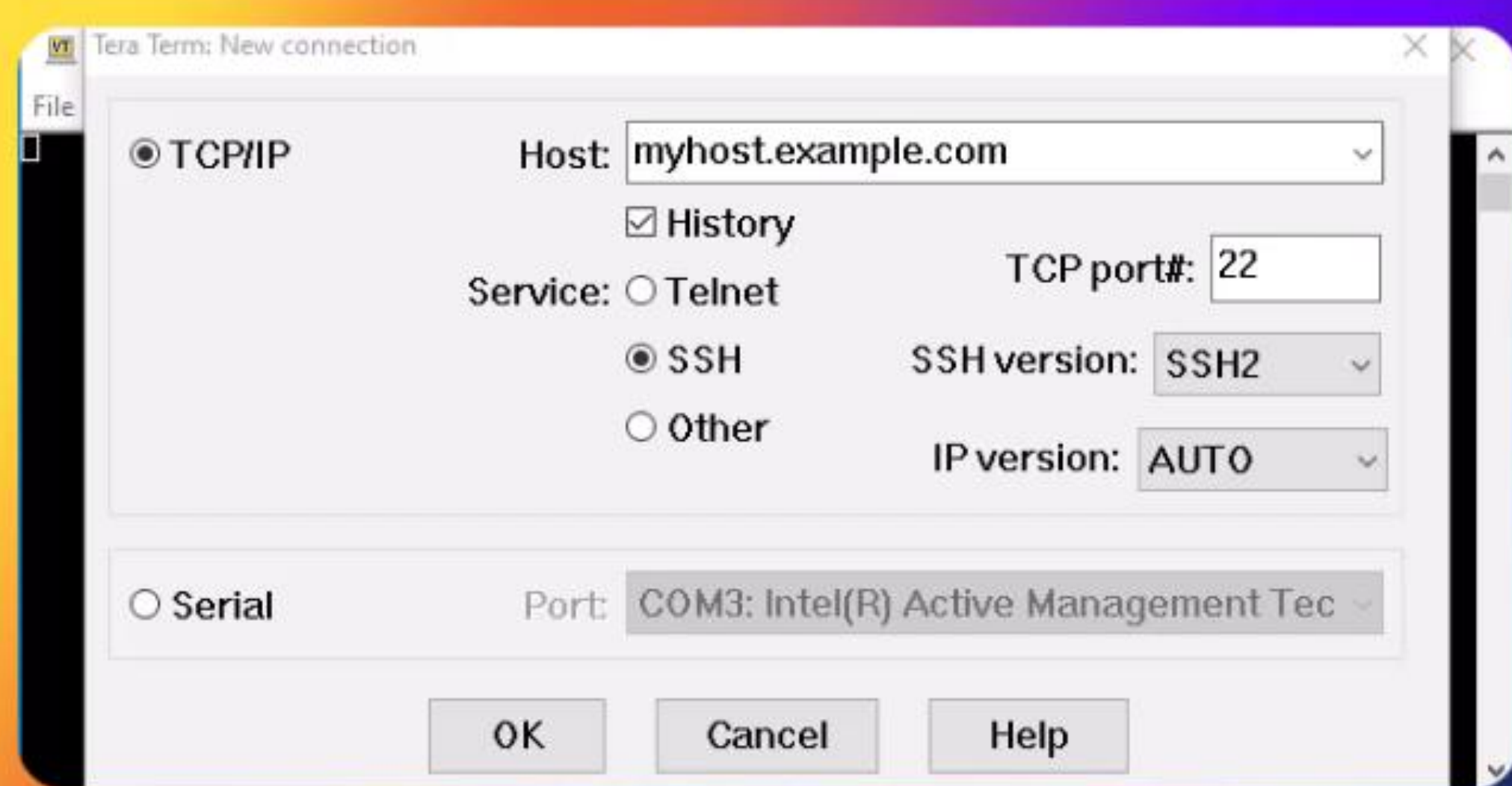
SOFTWARE TOOLS – APCFGBUILDER.EXE

Introduction

- Current ApCfgBuilder.exe tool (SW V1.4 and later)
 - Compatible with both versions of the CAN-USB converter
 - CAN-USB-COM converter
 - (green status LEDs)
 - CAN-USB-COM-FD converter
 - (blue status LEDs)
- Build 03.02.2021.13.21.42 in title bar



SETTING UP THE CAN CONVERTER



Tera Term: New connection

File

☒ TCP/IP

Host: myhost.example.com

☒ History

Service: ☐ Telnet

☒ SSH

☐ Other

TCP port#: 22

SSH version: SSH2

IP version: AUTO

☐ Serial

Port: COM3: Intel(R) Active Management Tec

OK Cancel Help

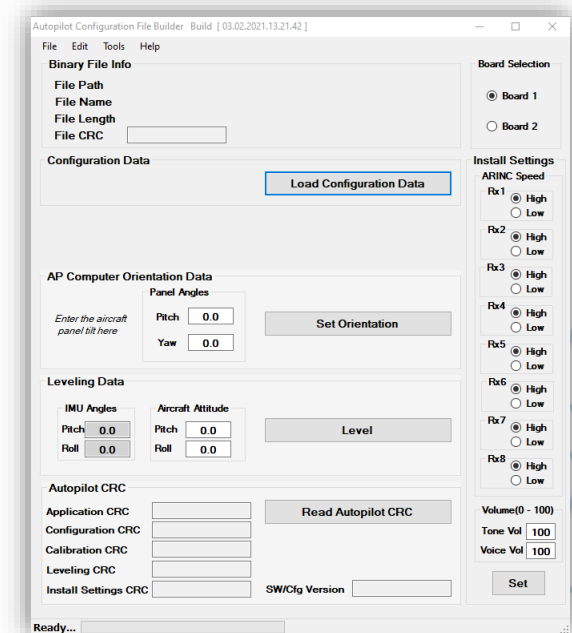
SOFTWARE TOOLS – APCFGBUILDER.EXE

Introduction

The ApCfgBuilder.exe application establishes a two-way communication between the PC and the FGC using the CAN-USB converter connected to the CAN 2 (Internal) maintenance connector.

The ApCfgBuilder.exe application (ApCfgBuilder) is used for the following purposes:

- ARINC 429 RX port speed configuration
- Voice and Tone volume setting
- IMU calibration
- Reading the CRCs from the FGC
- Updating or changing the configuration file



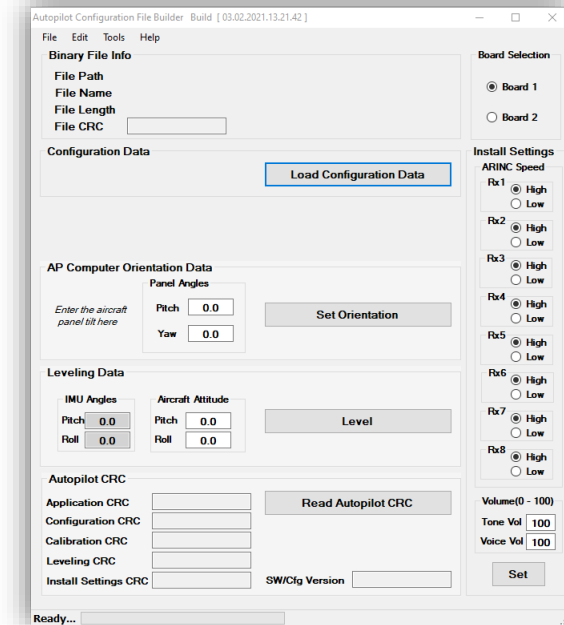
SOFTWARE TOOLS – APCFGBUILDER.EXE



Introduction

ApCfgBuilder will not:

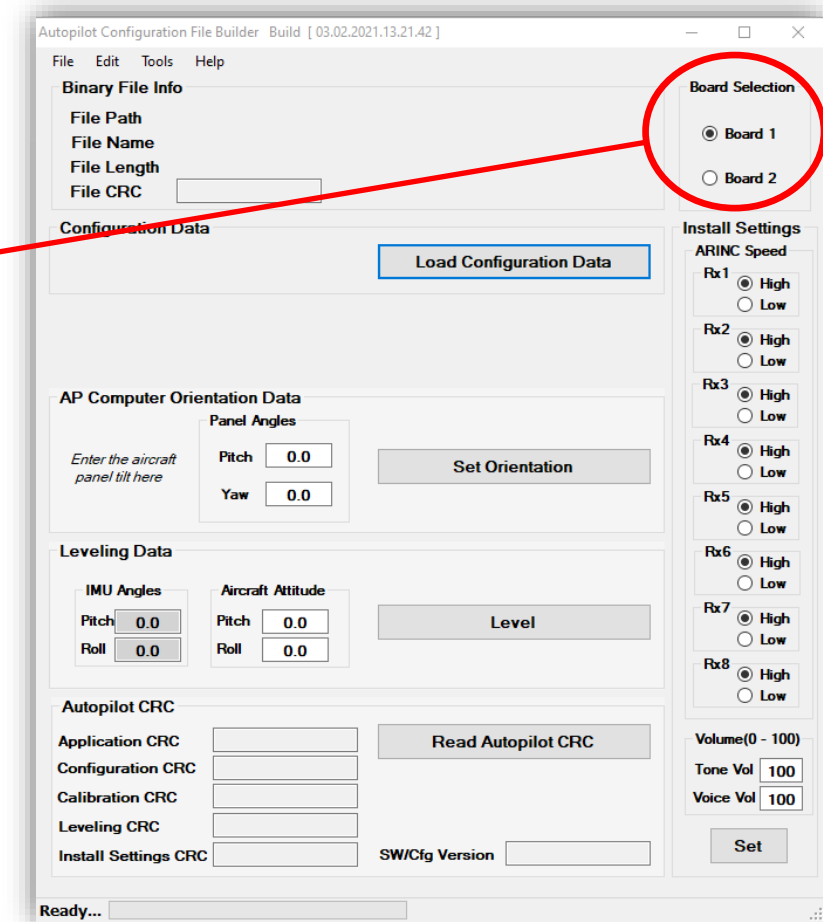
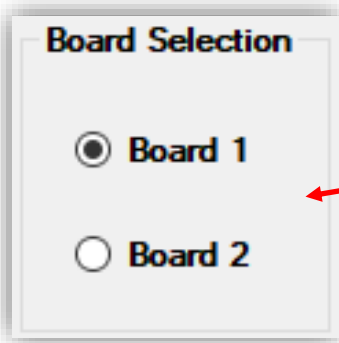
- Detect and display previously entered Panel Angles or Aircraft Attitude
- Detect and display ARINC 429 speeds already set on the FGC
- Detect and display Volume Settings



SOFTWARE TOOLS – APCFGBUILDER.EXE

Introduction

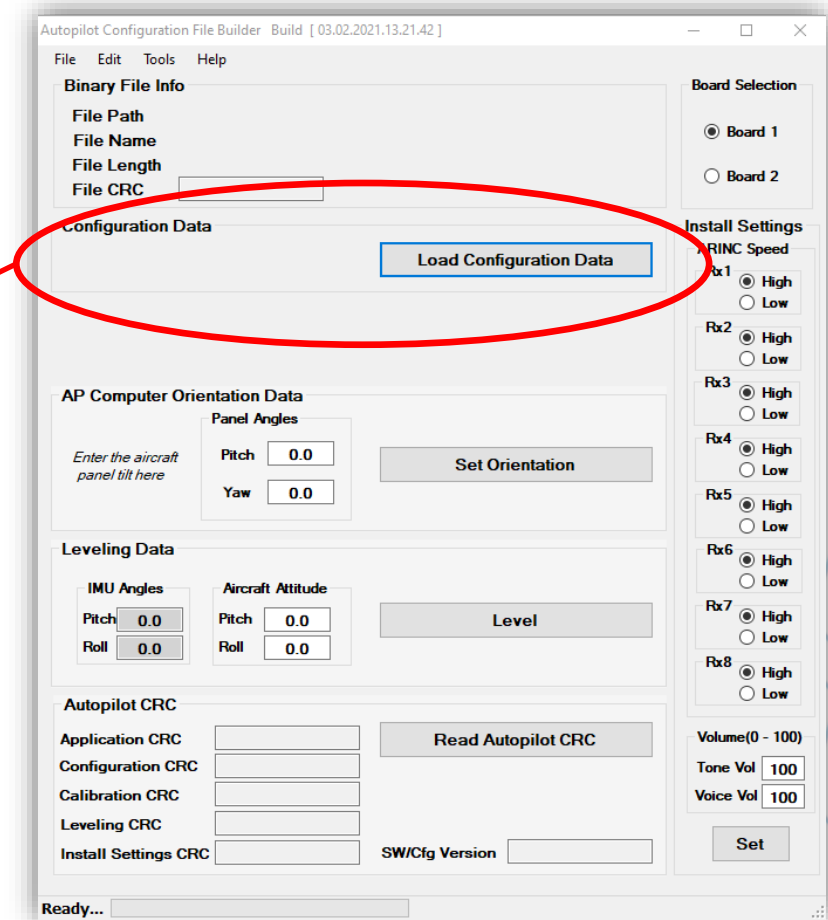
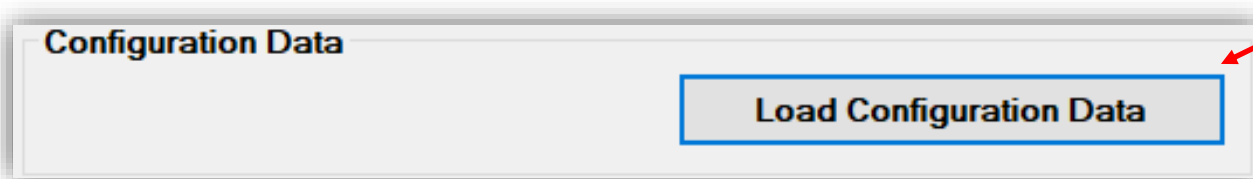
- Board Selection area
 - Selects which board the ApCfgBuilder will interact with



SOFTWARE TOOLS – APCFGBUILDER.EXE

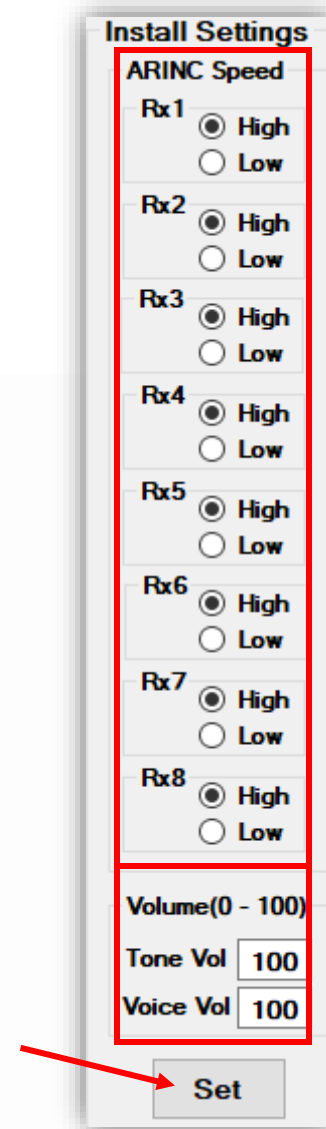
Introduction

- Configuration Data area
 - The Load Configuration Data button is used to load a config.bin file to the FGC via the CAN 2 maintenance connector



Introduction

- Install Settings area
 - ARINC Speed
 - Used to select the ARINC 429 receive speeds for the FGC
 - Volume (0 – 100)
 - Used to set the Tone and Voice volume levels
 - Levels should be set to similar values
- Set button
 - Creates an install.bin file with all of the selections in the ARINC Speed and Volume fields, then writes the file to the FGC.
 - Does not affect IMU calibration



Install Settings

ARINC Speed

Rx1 ☒ High ☐ Low

Rx2 ☒ High ☐ Low

Rx3 ☒ High ☐ Low

Rx4 ☒ High ☐ Low

Rx5 ☒ High ☐ Low

Rx6 ☒ High ☐ Low

Rx7 ☒ High ☐ Low

Rx8 ☒ High ☐ Low

Volume(0 - 100)

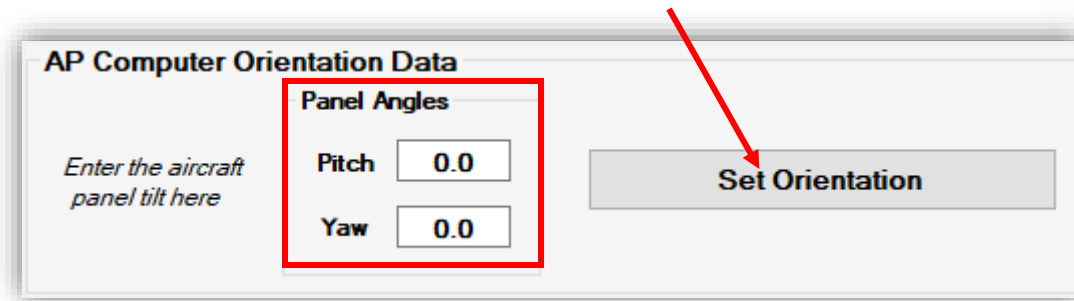
Tone Vol

Voice Vol

Set

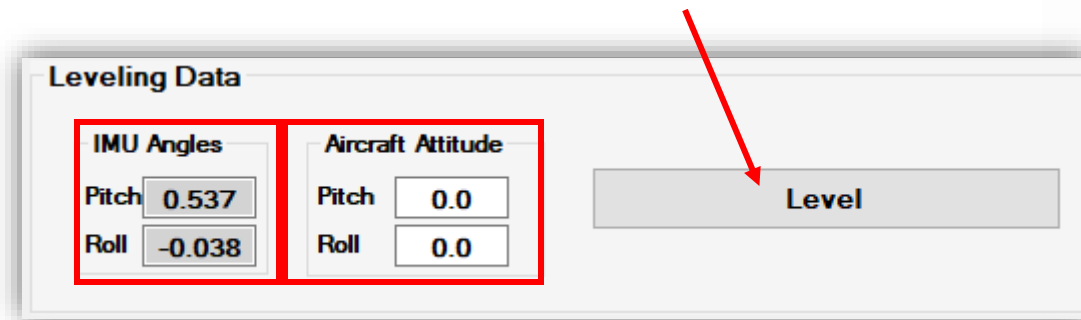
Introduction

- AP Computer Orientation Data area
 - Panel Angles - Enter the Pitch and Yaw values for the orientation of the FGC as installed in the radio or instrument panel
 - Set Orientation Button – creates a level.bin file with the Panel Angle Pitch and Yaw values, then writes the file to the FGC
 - Be sure to set the AP Computer Orientation Data before leveling the IMU
 - Must be set each time IMU calibration is performed
 - Does not affect install settings



Introduction

- Leveling Data area
 - IMU Angles
 - Displays Pitch and Roll values for the selected IMU in real time
 - These are the values that are calibrated
 - Aircraft Attitude
 - Enter the measured aircraft attitude for Pitch and Roll
 - These are the values the selected IMU is being calibrated to
- Level button
 - Creates a level.bin file and loads it to the FGC
 - Set the Computer Orientation prior to using the Level Button
 - Does not affect install settings



Introduction

What is a CRC?

Cyclic redundancy check

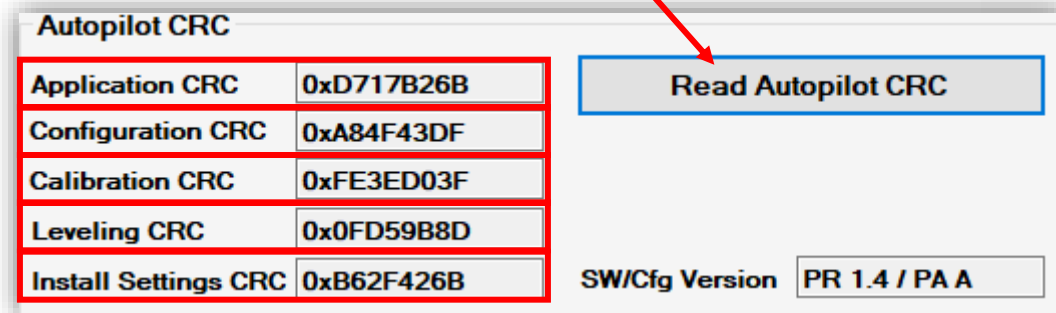
From Wikipedia, the free encyclopedia

A **cyclic redundancy check (CRC)** is an error-detecting code commonly used in digital networks and storage devices to detect accidental changes to digital data. Blocks of data entering these systems get a short *check value* attached, based on the remainder of a polynomial division of their contents. On retrieval, the calculation is repeated and, in the event the check values do not match, corrective action can be taken against data corruption. CRCs can be used for error correction.

CRCs are so called because the *check* (data verification) value is a *redundancy* (it expands the message without adding information) and the algorithm is based on *cyclic* codes. CRCs are popular because they are simple to implement in binary hardware, easy to analyze mathematically, and particularly good at detecting common errors caused by noise in transmission channels. Because the check value has a fixed length, the function that generates it is occasionally used as a hash function.

Introduction

- Autopilot CRC area
 - Read Autopilot CRC button requests the CRC values from the FGC
 - Application CRC – Board 1 and Board 2 must match
 - Configuration CRC – Board 1 and Board 2 must match
 - Calibration CRC - Each board may have unique values
 - Leveling CRC – Each board may have unique values
 - Install Settings CRC – Board 1 and Board 2 must match



The screenshot shows a dialog box titled "Autopilot CRC". It contains a table with five rows of CRC values, each row highlighted with a red border. To the right of the table is a button labeled "Read Autopilot CRC", which is highlighted with a blue border and pointed to by a red arrow. Below the table and button is a label "SW/Cfg Version" followed by a text box containing "PR 1.4 / PA A".

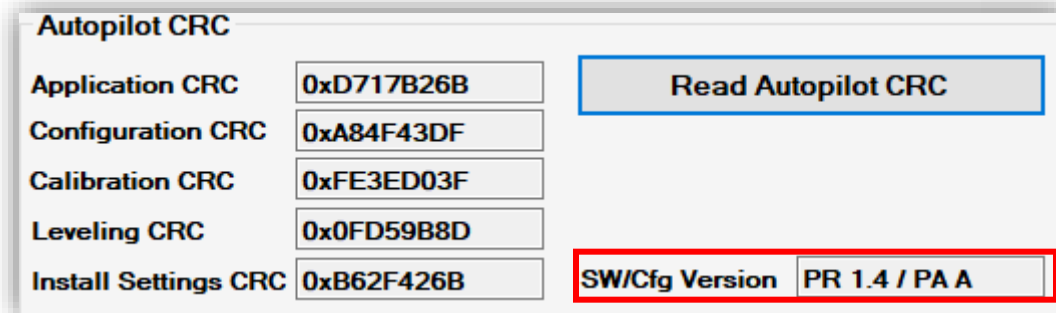
Autopilot CRC	
Application CRC	0xD717B26B
Configuration CRC	0xA84F43DF
Calibration CRC	0xFE3ED03F
Leveling CRC	0x0FD59B8D
Install Settings CRC	0xB62F426B

Read Autopilot CRC

SW/Cfg Version PR 1.4 / PA A

Introduction

- Autopilot CRC area
 - SW/Cfg Version – displays the SW version and basic configuration file version information.
 - This is not specific enough to determine exact SW or configuration file version.



The screenshot shows a dialog box titled "Autopilot CRC". It contains a table of CRC values and a "Read Autopilot CRC" button. The "SW/Cfg Version" field is highlighted with a red border.

Autopilot CRC	
Application CRC	0xD717B26B
Configuration CRC	0xA84F43DF
Calibration CRC	0xFE3ED03F
Leveling CRC	0x0FD59B8D
Install Settings CRC	0xB62F426B

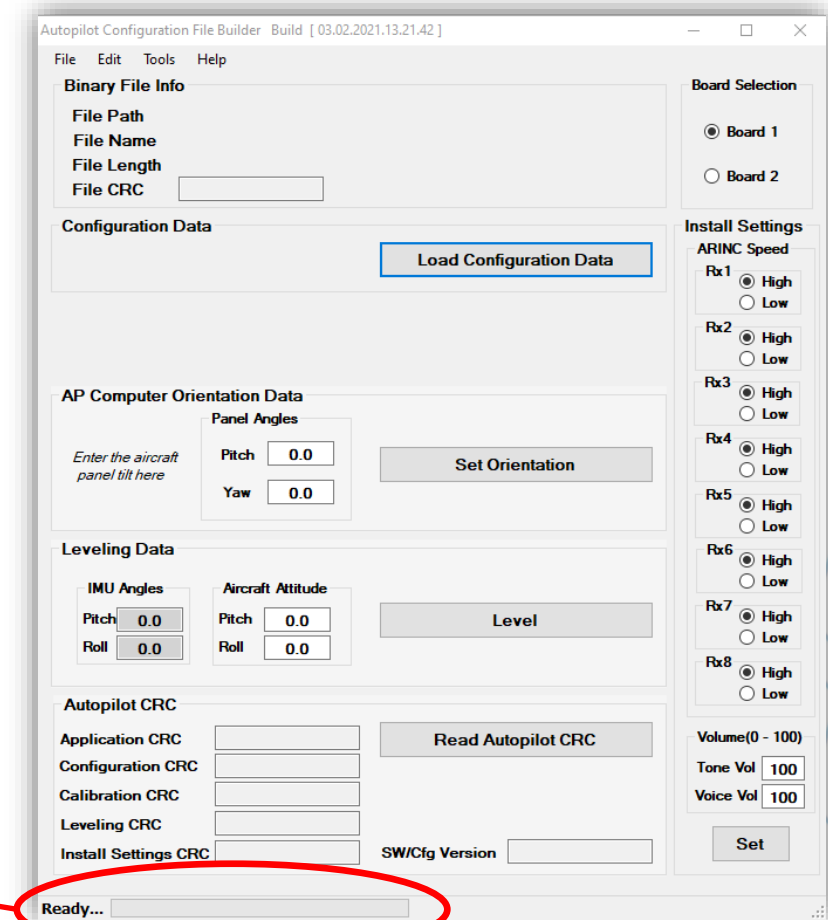
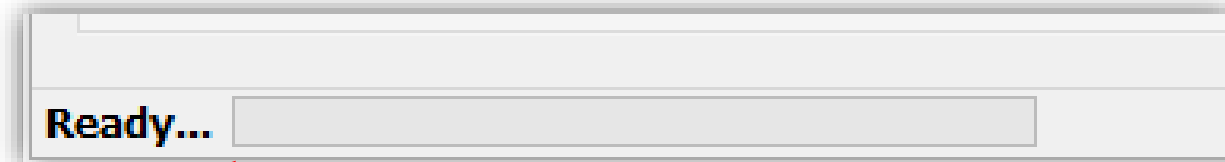
Read Autopilot CRC

SW/Cfg Version PR 1.4 / PA A

SOFTWARE TOOLS – APCFGBUILDER.EXE

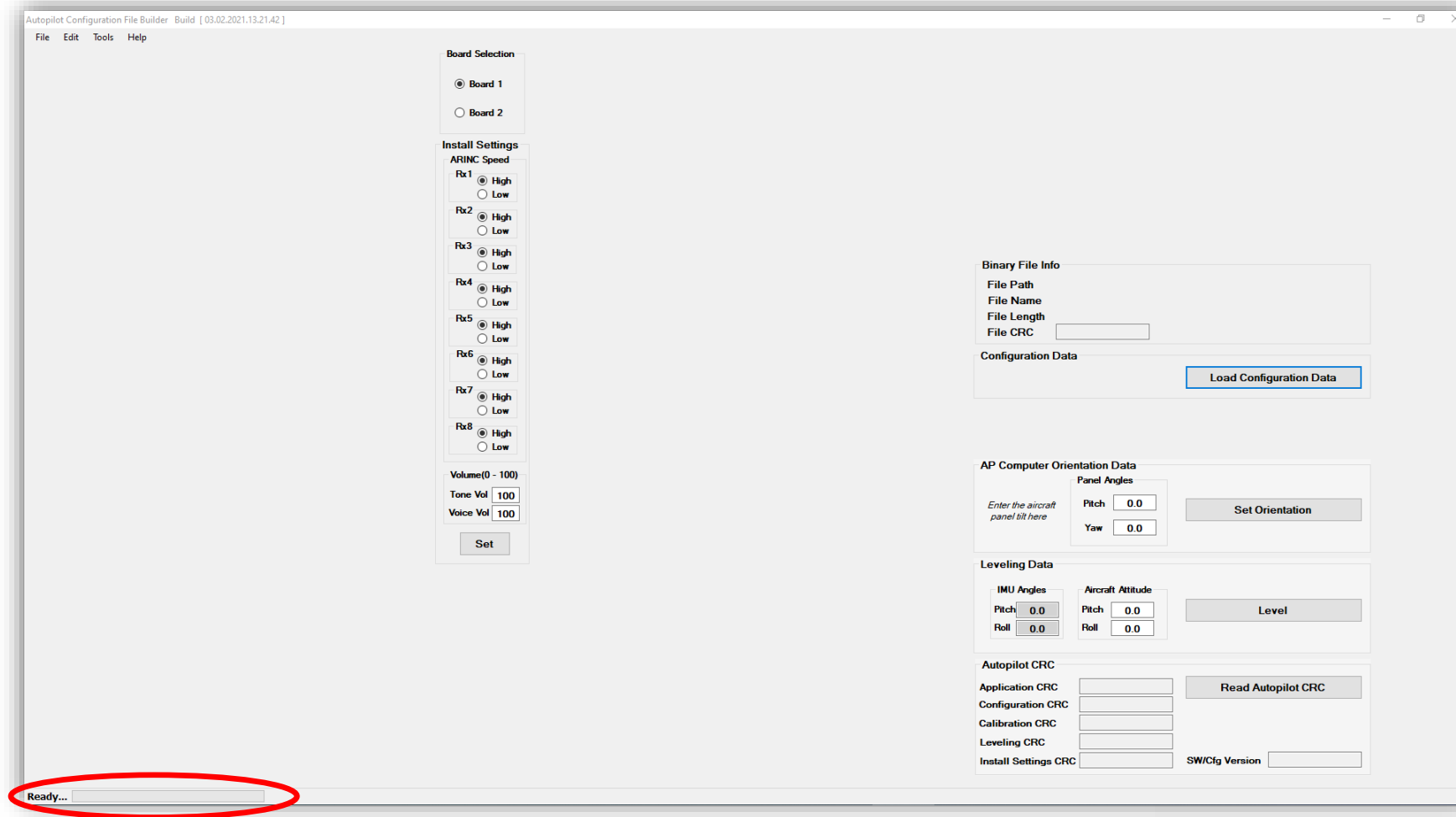
Introduction

- Status bar
 - Displays status of last interaction
 - Lower resolution displays may require maximizing the window to view the Status Bar



SOFTWARE TOOLS – APCFGBUILDER.EXE

Introduction



Autopilot Configuration File Builder Build [03.02.2021.13.21.42]

File Edit Tools Help

Board Selection

☒ Board 1
☐ Board 2

Install Settings

ARINC Speed

Rx1 ☒ High ☐ Low
Rx2 ☒ High ☐ Low
Rx3 ☒ High ☐ Low
Rx4 ☒ High ☐ Low
Rx5 ☒ High ☐ Low
Rx6 ☒ High ☐ Low
Rx7 ☒ High ☐ Low
Rx8 ☒ High ☐ Low

Volume(0 - 100)

Tone Vol
Voice Vol

Set

Binary File Info

File Path
File Name
File Length
File CRC

Configuration Data

Load Configuration Data

AP Computer Orientation Data

Panel Angles

Enter the aircraft panel tilt here

Pitch
Yaw

Set Orientation

Leveling Data

IMU Angles

Pitch
Roll

Aircraft Attitude

Pitch
Roll

Level

Autopilot CRC

Application CRC
Configuration CRC
Calibration CRC
Leveling CRC
Install Settings CRC

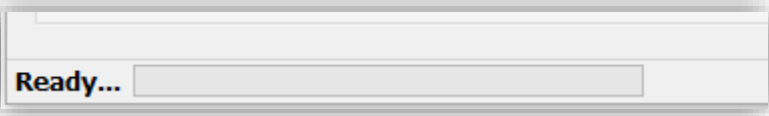
Read Autopilot CRC

SW/Cfg Version

Ready...

Introduction

- Status bar



Ready...

Good

Successfully Read CRCs from AP

Leveling Data Successfully Loaded to AP

Install Settings Successfully Loaded to AP

Bad

Failed to Read CRC from AP

Failed to Load Leveling Data to AP

Failed to Load Install Settings Data to AP

Required equipment

- A PC or Laptop with Windows 7 or later is required.
 - Must have an available USB port



Required equipment

- A PC or Laptop with Windows 7 or later is required.
 - Must have an available USB port
- An appropriate CAN-USB converter. Refer to the ICA for compatible converters.



gridconnect
CAN-USB-COM



OR

gridconnect
CAN-USB-COM-FD



Required equipment

- A PC or Laptop with Windows 7 or later is required.
 - Must have an available USB port
- An appropriate CAN-USB converter. Refer to the ICA for compatible converters.
 - DO NOT use a USB extender. The converter must be plugged directly into the laptop



TROUBLESHOOTING ATTITUDE FAIL



TROUBLESHOOTING ATTITUDE FAIL



TROUBLESHOOTING ATTITUDE FAIL



Troubleshooting an ATTITUDE FAIL message

The 3100 requires ARINC 429 Label 210 (L210) True Air Speed (TAS) to initialize the IMU(s). If this is not received or is not valid, it will display ATTITUDE FAIL.

- Missing ARINC 429 L210 – True Air Speed (TAS)
 - Check and resolve:
 - ADC powered on and valid
 - Bad or missing OAT probe
 - ADC requires OAT probe to provide valid TAS
 - Failed ADC



TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message

- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - ADC correctly configured
 - Garmin EFIS with integrated ADC and AHRS require General purpose IO to be configured for ADC and AHRS
 - Refer to S-Tec document 87338



GDU 1 General Purpose A429 Out Settings		
	Port	Speed
General Purpose 1 #1	None	Low
General Purpose 1 #2	None	Low
General Purpose 2 #1	None	Low
General Purpose 2 #2	None	Low
Integrated AHRS	Port 2	High
Integrated ADC	Port 3	Low
Integrated ADAHRS	None	High

TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message

- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - ARINC 429 wiring between ADC and FGC
 - Wire reversed
 - grounded/shorted
 - open

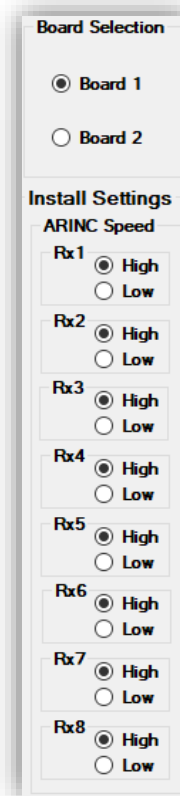


TROUBLESHOOTING ATTITUDE FAIL



Troubleshooting an ATTITUDE FAIL message

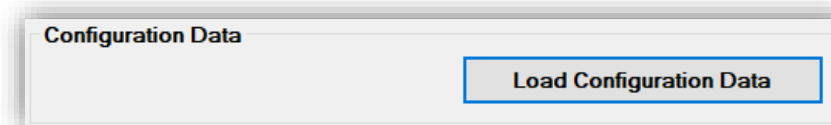
- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - FGC ARINC 429 RX speed must match ADC source TX speed
 - Aspen ProMax w/SW V2.11 – HIGH speed
 - Aspen EFD/ProMax pre SW V2.11 – either HIGH or LOW speed
 - Match the Aspen EFD1000 configuration
 - Garmin ADC – LOW speed
 - Sandia ADC – HIGH speed



TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message

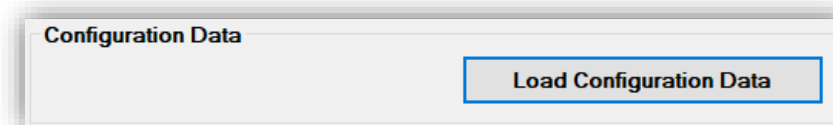
- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - Is the correct configuration file loaded?
 - Config 1 – ADC ARINC 429 is wired to RX 5 and RX 6
 - All other configurations – ADC data wired to RX2
 - Refer to document 87338 and the appropriate wiring diagram



TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message

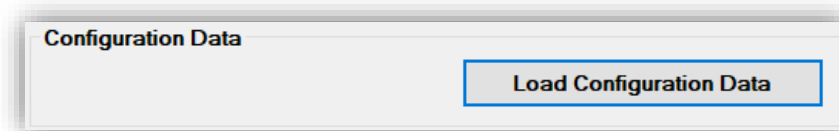
- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - Is the correct configuration file loaded?
 - Common cause: Upgrading Aspen interface to ProMax w/ Advanced Autopilot Unlock changes from Config 1 to Config 2



TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message

- Missing ARINC 429 L210 – True Air Speed
 - Check and resolve:
 - Is the correct configuration file loaded?
 - Compare configuration file P/N to STC parts list
 - Reference S-Tec wiring diagram and 87338 document
 - Check Configuration CRC
 - Contact Technical Support for assistance



TROUBLESHOOTING ATTITUDE FAIL

Troubleshooting an ATTITUDE FAIL message



Questions?

TROUBLESHOOTING AP FAIL (POST)



TROUBLESHOOTING AP FAIL (POST)



TROUBLESHOOTING AP FAIL (POST)



TROUBLESHOOTING AP FAIL (POST)



TROUBLESHOOTING AP FAIL (POST)



Now what??!!

TROUBLESHOOTING AP FAIL (POST)

AP FAIL

- An AP FAIL indicates the FGC has detected an internal or external failure of the DFCS
 - The AP, FD, and YD cannot be engaged until the AP FAIL is resolved
 - Manual Electric Trim may operate, depending on the cause of the AP FAIL



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL message



Bad or missing ARINC 429 data will not cause an AP FAIL message

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL message

- AP FAIL after Power On Self Test (POST) can happen for a number of reasons
 - One or more bad or disconnected servo(s)
 - Bad or corrupt IMU calibration
 - Incorrect 28VDC/14VDC strapping
 - Missing power input
 - Mismatched Configuration files on board 1 and board 2
 - Missing configuration file
 - Wrong configuration file for FGC SW version
 - Mismatched ARINC 429 speed settings on board 1 and board 2
 - Internal FGC failure
 - Etc.



Where do you start?

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL message

- Typical method:
 - Grab a multimeter
 - Reference wiring diagram
 - Start 'ringing out the wiring'
 - Potentially spend hours 'checking everything'
 - Change the FGC if you can get a spare
 - Change one or more of the servos if you can get spares
 - Call technical support (after spending hours 'checking everything')
 - Success through brute force troubleshooting



Troubleshooting an AP FAIL message

- Another method:
 - Use a PC and CAN-USB converter (already in hand) with TSOVista for a more direct troubleshooting approach.

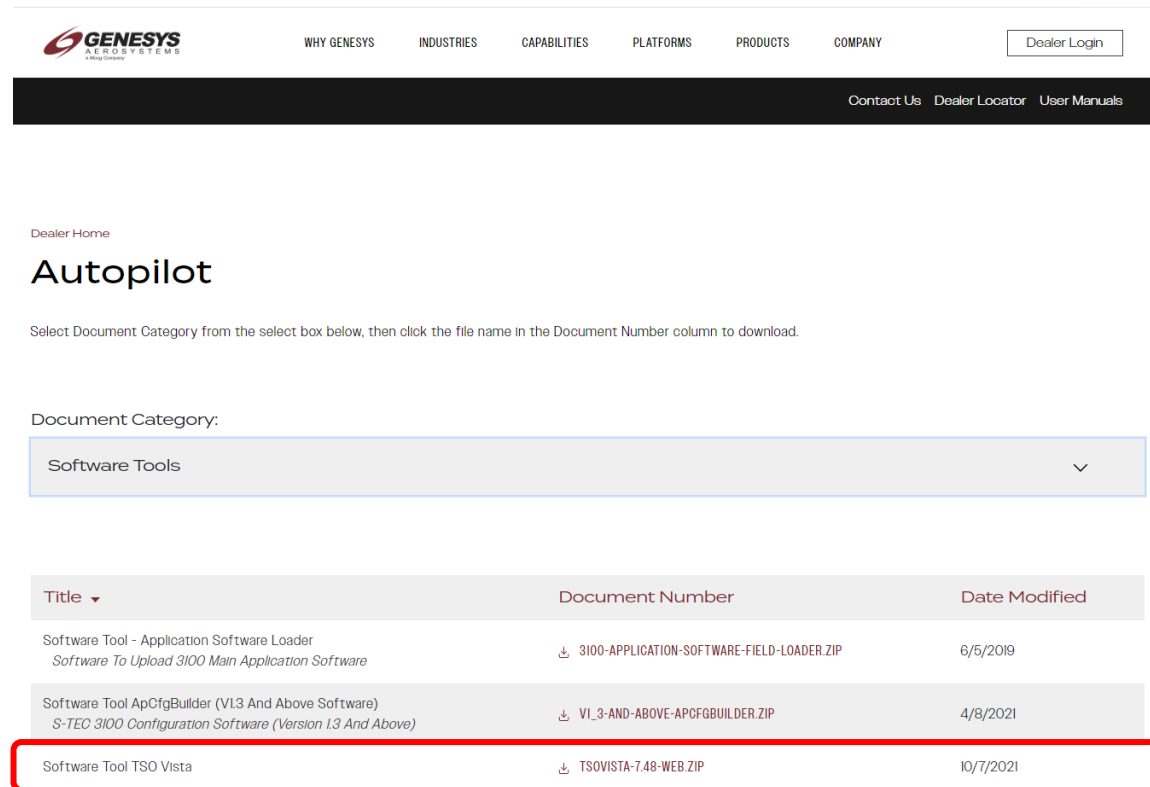


We will come back to resolving the **AP FAIL** shortly.

SOFTWARE TOOLS – TSOVISTA

TSOVista Introduction

- TSOVista can be downloaded from the dealer section of the website:
 - Home>Dealer>Autopilot Documents>Installation Documents
- Installation instructions are included with the downloaded zip file.
- If you have an older version of TSOVista, it is a good idea to remove it and any associated files and folders before installing the current version



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Autopilot

Select Document Category from the select box below, then click the file name in the Document Number column to download.

Document Category:

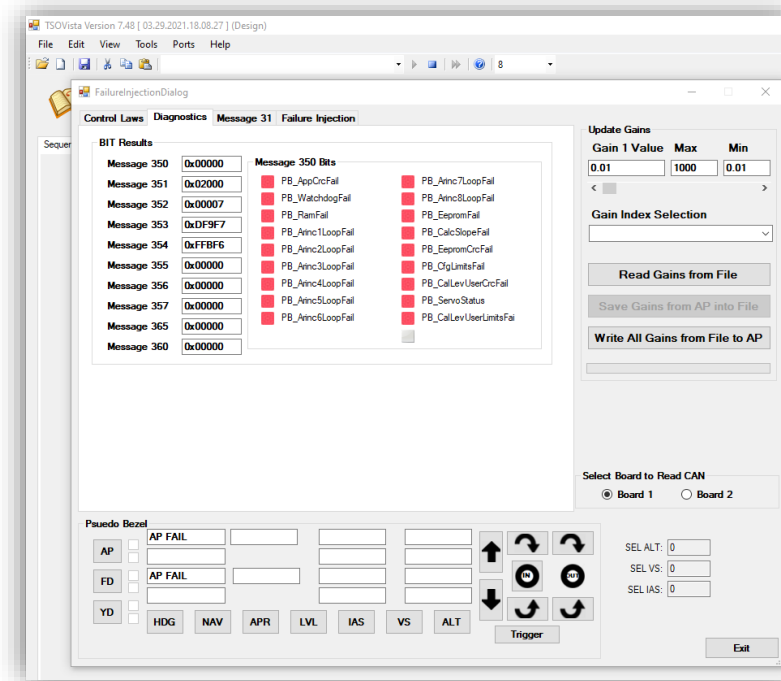
Software Tools

Title ▼	Document Number	Date Modified
Software Tool - Application Software Loader <i>Software To Upload 3100 Main Application Software</i>	3100-APPLICATION-SOFTWARE-FIELD-LOADER.ZIP	6/5/2019
Software Tool ApCfgBuilder (V1.3 And Above Software) <i>S-TEC 3100 Configuration Software (Version 1.3 And Above)</i>	V1_3-AND-ABOVE-APCFGBUILDER.ZIP	4/8/2021
Software Tool TSO Vista	TSOVISTA-7.48-WEB.ZIP	10/7/2021

TSOVista Introduction

The TSOVista application can be used to help monitor and diagnose issues with the S-TEC 3100 autopilot.

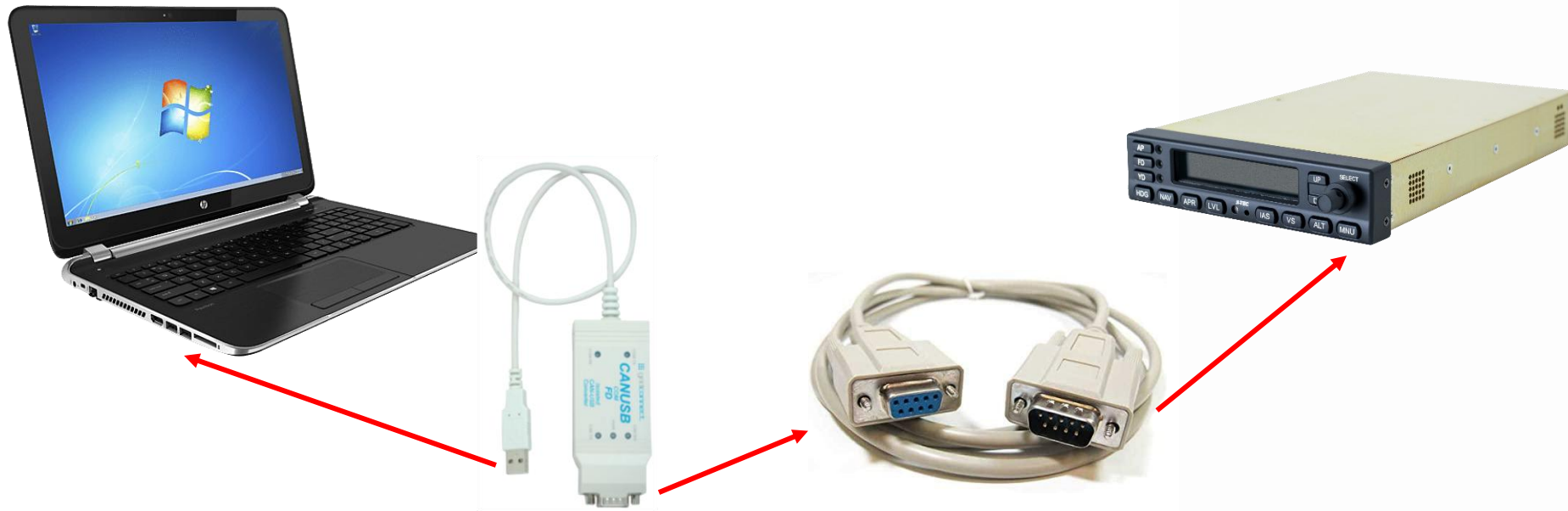
- TSOVista uses the CAN-USB Converter to listen to the CAN bus messages from the FGC.
- The data on the CAN bus is converted to readable data which can be used for diagnostics and recording.



TSOVista Introduction

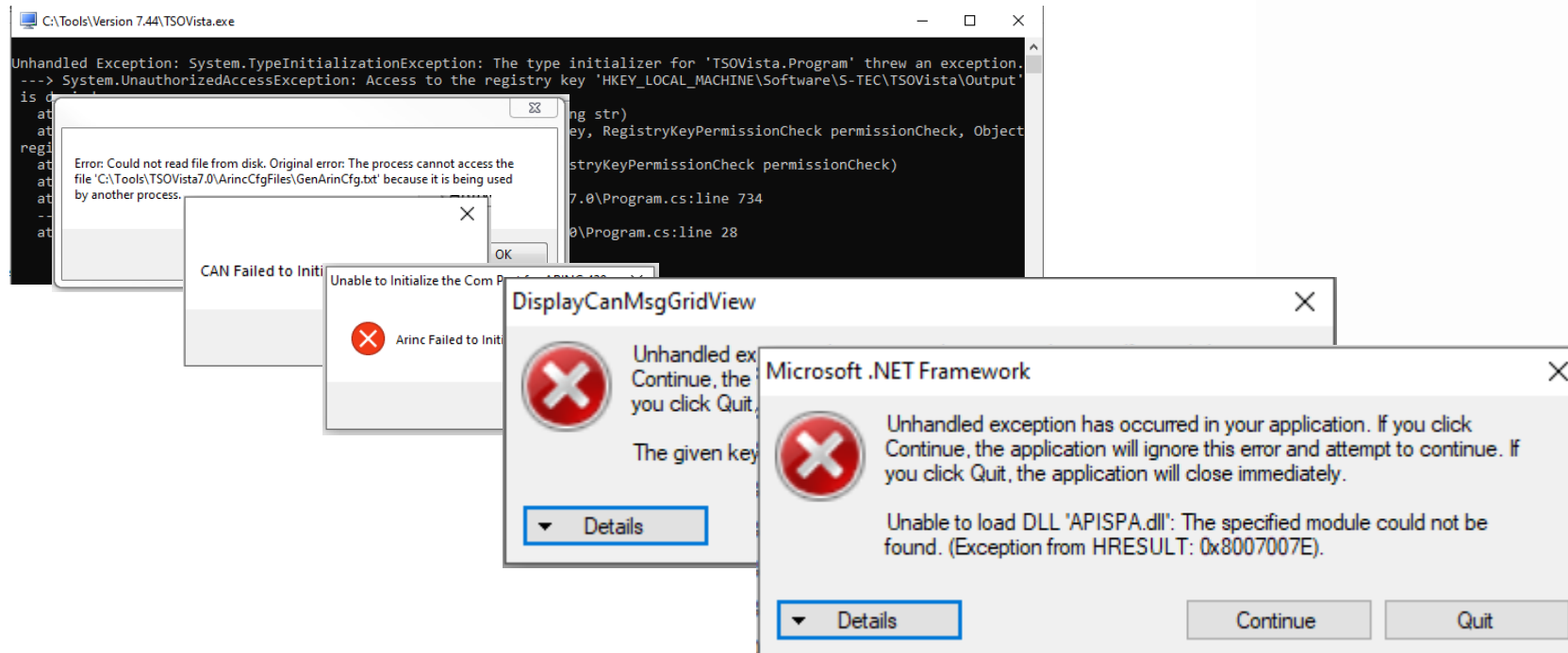
TSOVista Requirements

- PC or laptop with Windows 7 OS or later
- The same CAN-USB converter used with ApCfgBuilder
 - The CAN-USB converter must be assigned to COM40



TSOVista Introduction

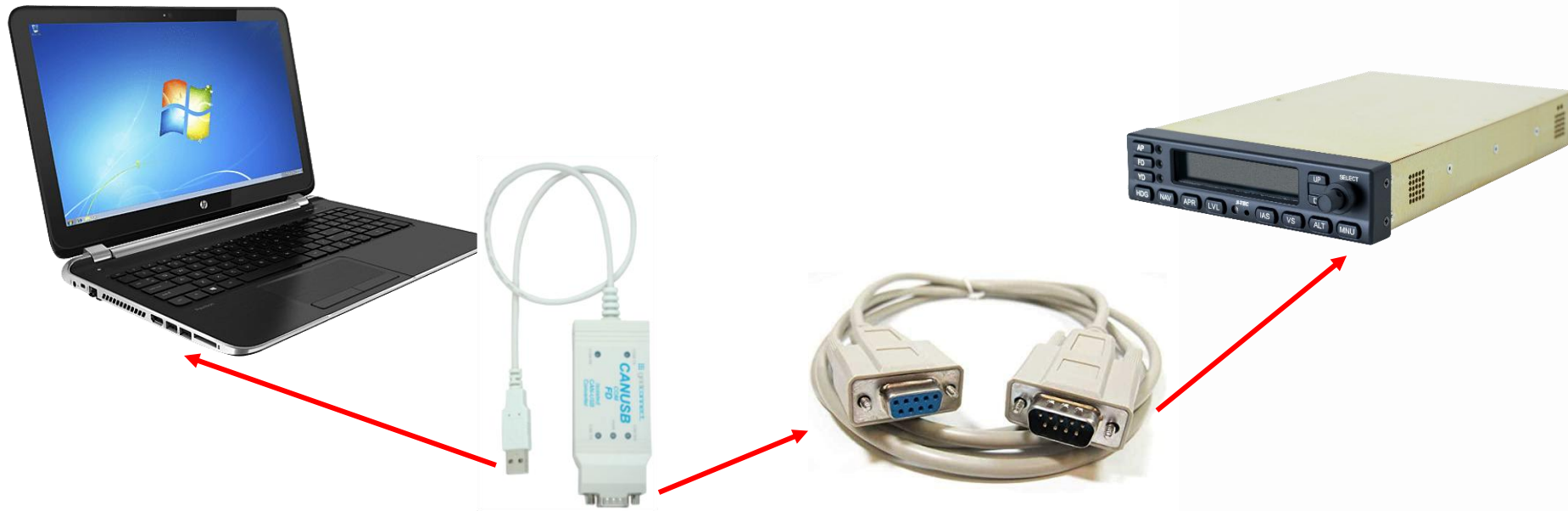
- To avoid confusion and wasted effort, be sure to follow the setup instructions in the document closely.
- TSOVista Error messages are covered in the end of the document.



TSOVista Introduction

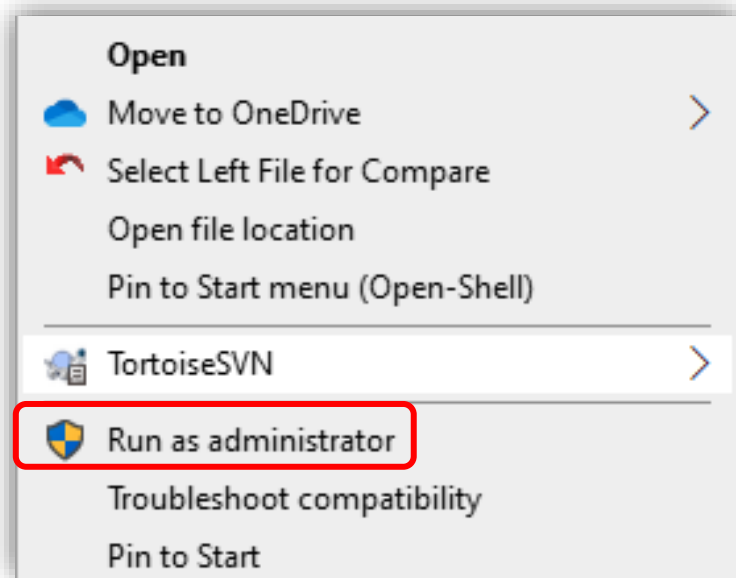
Once TSOVista is installed, it can be used for troubleshooting.

- First connect the CAN-USB converter to a USB port on the laptop
- Then connect the CAN-USB converter to the CAN 2 (internal) maintenance connector
- Power on the 3100 DFCS (if not on already)



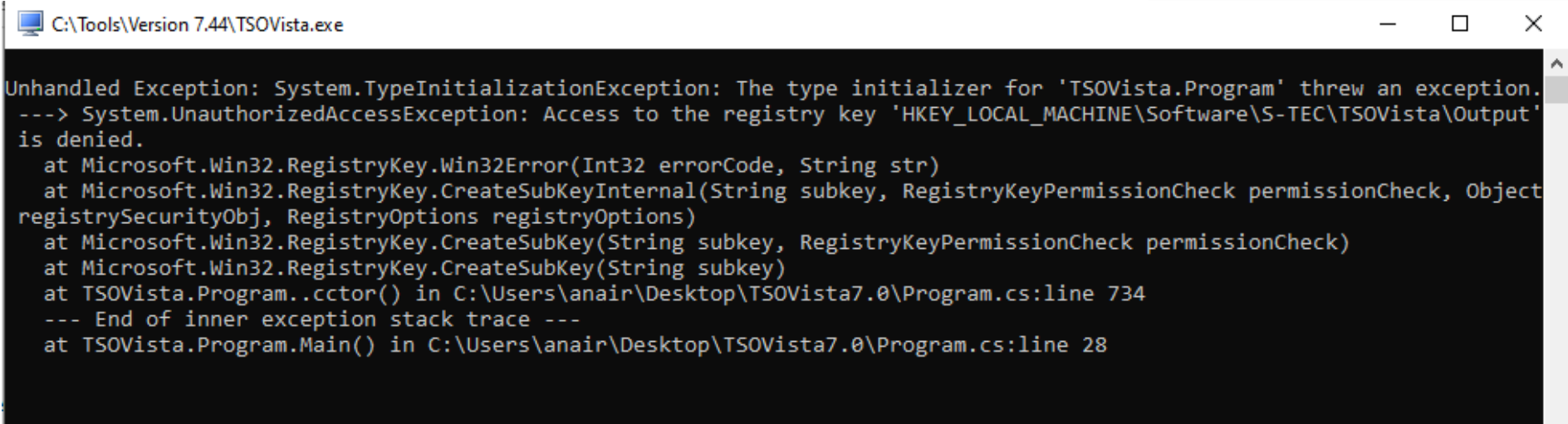
TSOVista Introduction

Run TSOVista as Administrator.



TSOVista Introduction

Run TSOVista as Administrator.

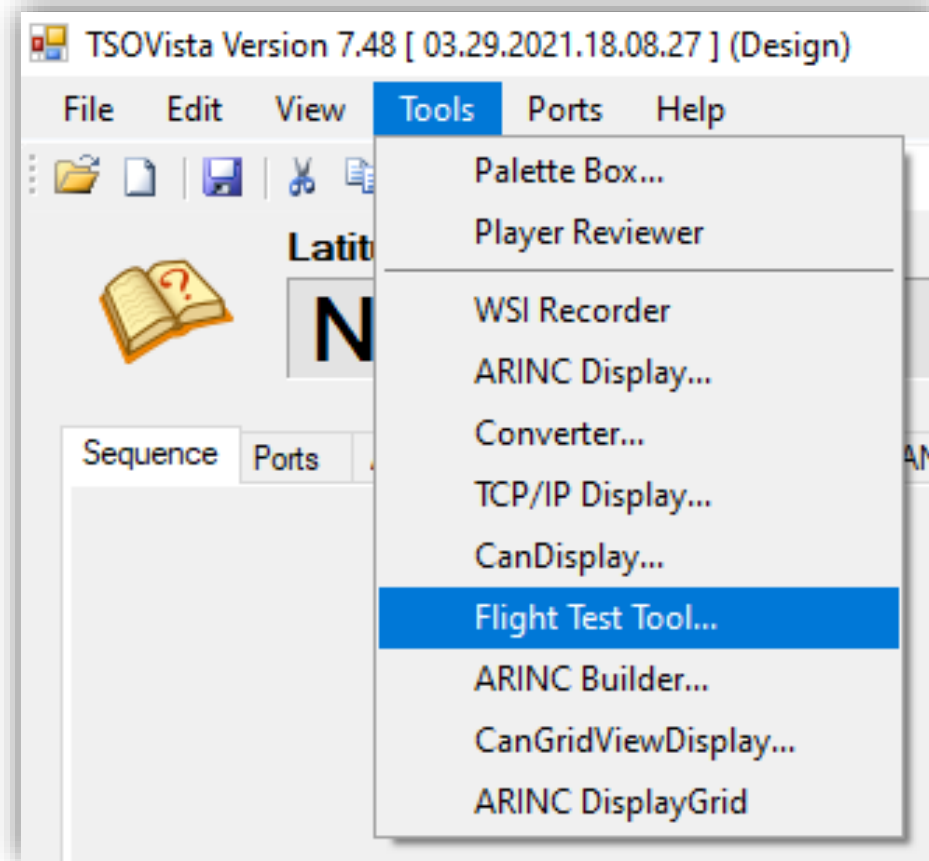


```
C:\Tools\Version 7.44\TSOVista.exe

Unhandled Exception: System.TypeInitializationException: The type initializer for 'TSOVista.Program' threw an exception.
---> System.UnauthorizedAccessException: Access to the registry key 'HKEY_LOCAL_MACHINE\Software\S-TEC\TSOVista\Output'
is denied.
   at Microsoft.Win32.RegistryKey.Win32Error(Int32 errorCode, String str)
   at Microsoft.Win32.RegistryKey.CreateSubKeyInternal(String subkey, RegistryKeyPermissionCheck permissionCheck, Object
registrySecurityObj, RegistryOptions registryOptions)
   at Microsoft.Win32.RegistryKey.CreateSubKey(String subkey, RegistryKeyPermissionCheck permissionCheck)
   at Microsoft.Win32.RegistryKey.CreateSubKey(String subkey)
   at TSOVista.Program..cctor() in C:\Users\anair\Desktop\TSOVista7.0\Program.cs:line 734
--- End of inner exception stack trace ---
   at TSOVista.Program.Main() in C:\Users\anair\Desktop\TSOVista7.0\Program.cs:line 28
```

TSOVista Introduction – Flight Test Tool

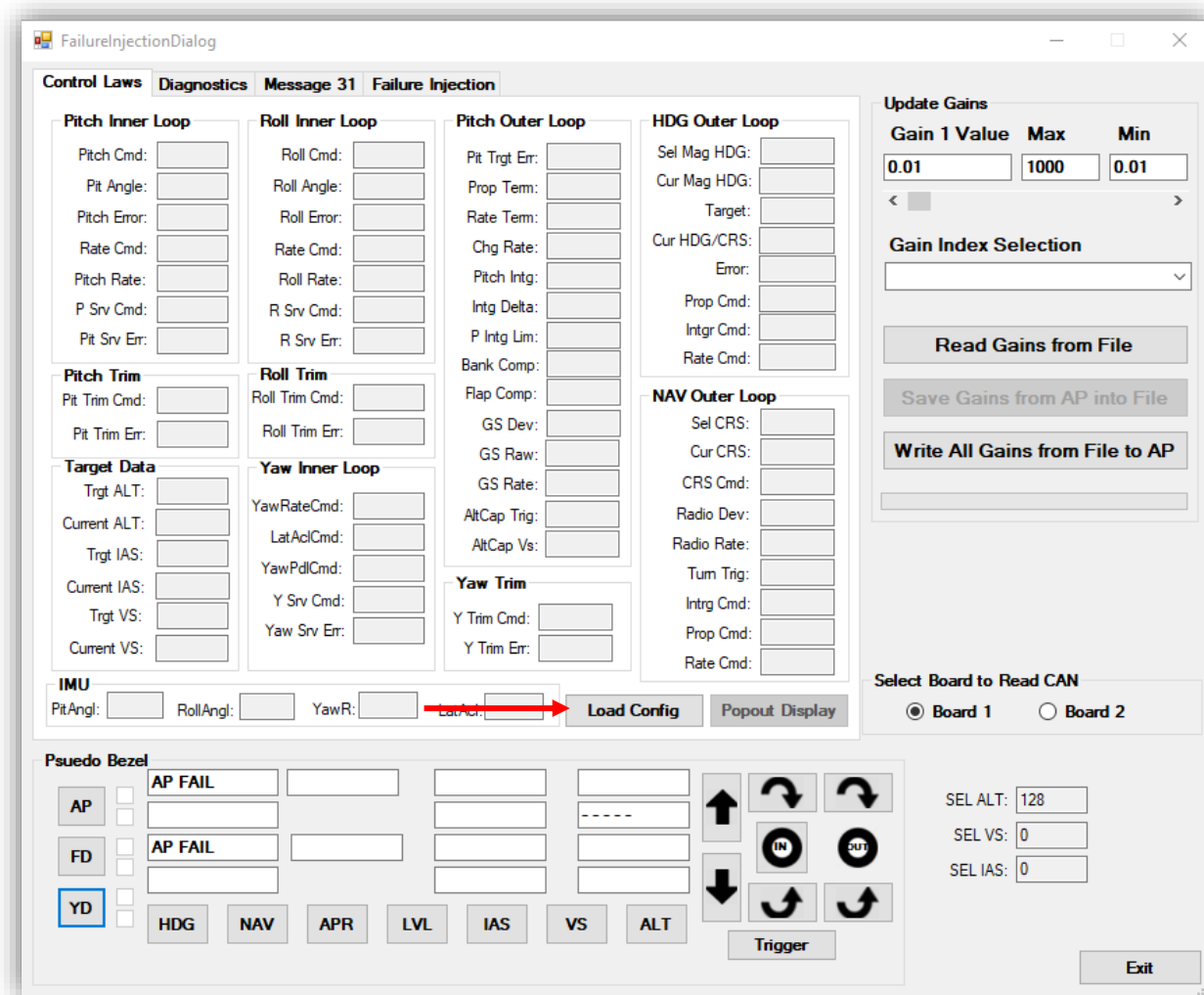
- From the Menu bar, select Tools>Flight Test Tool



SOFTWARE TOOLS – TSOVISTA

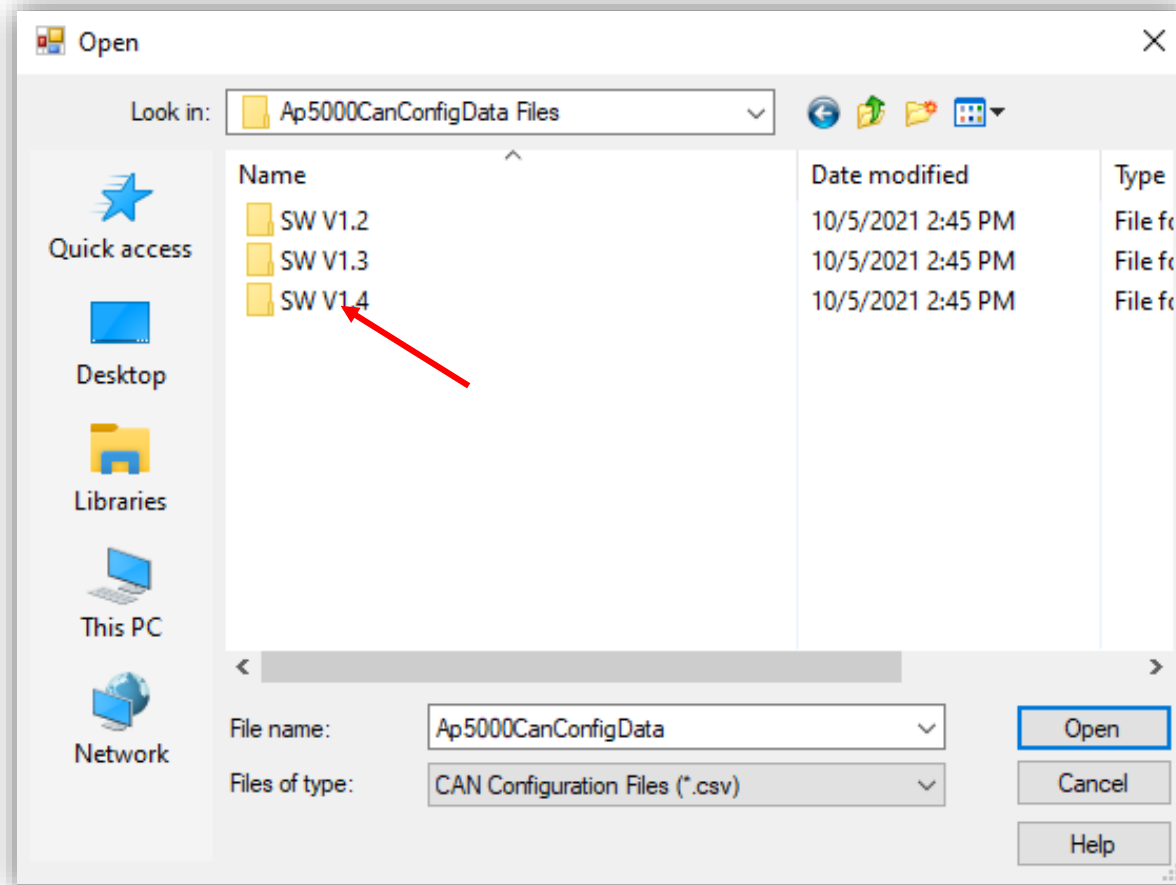
TSOVista Introduction – Flight Test Tool

- Click the Load Config button



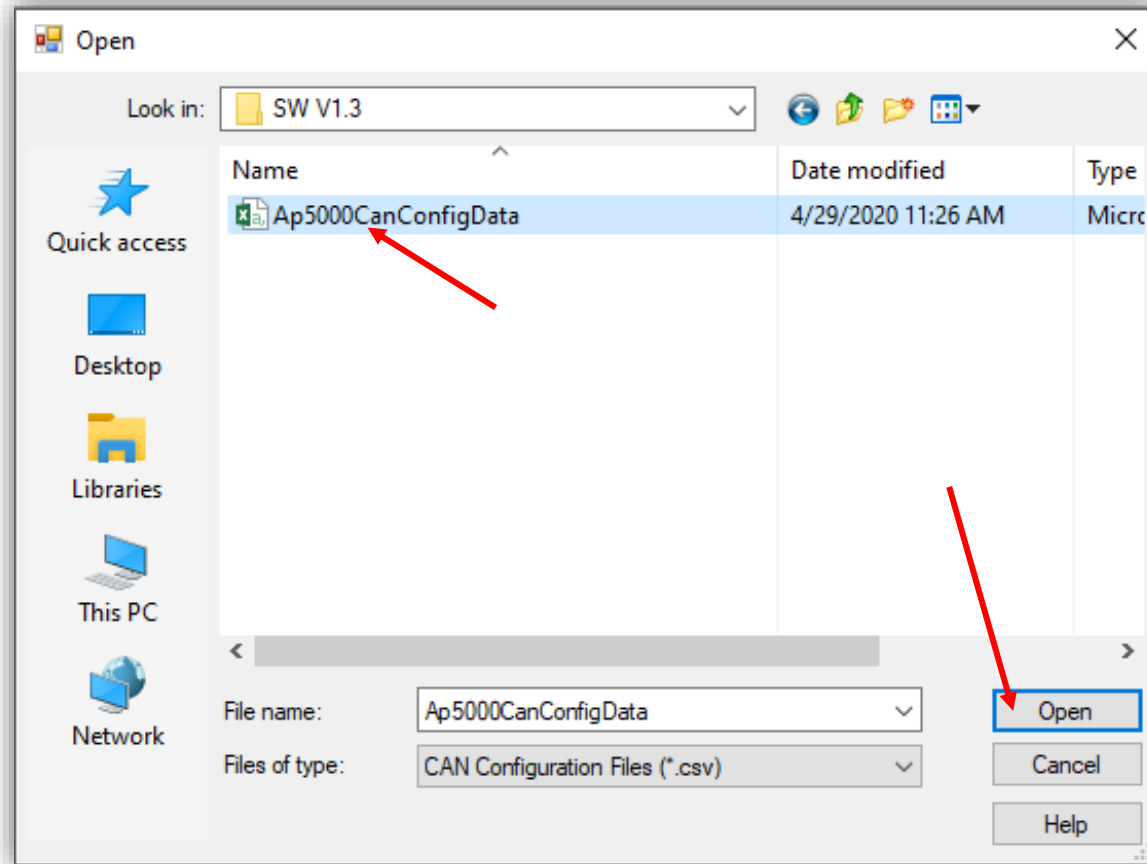
TSOVista Introduction – Flight Test Tool

- Open the SW 1.X folder that corresponds to the Application SW version on the FGC



TSOVista Introduction – Flight Test Tool

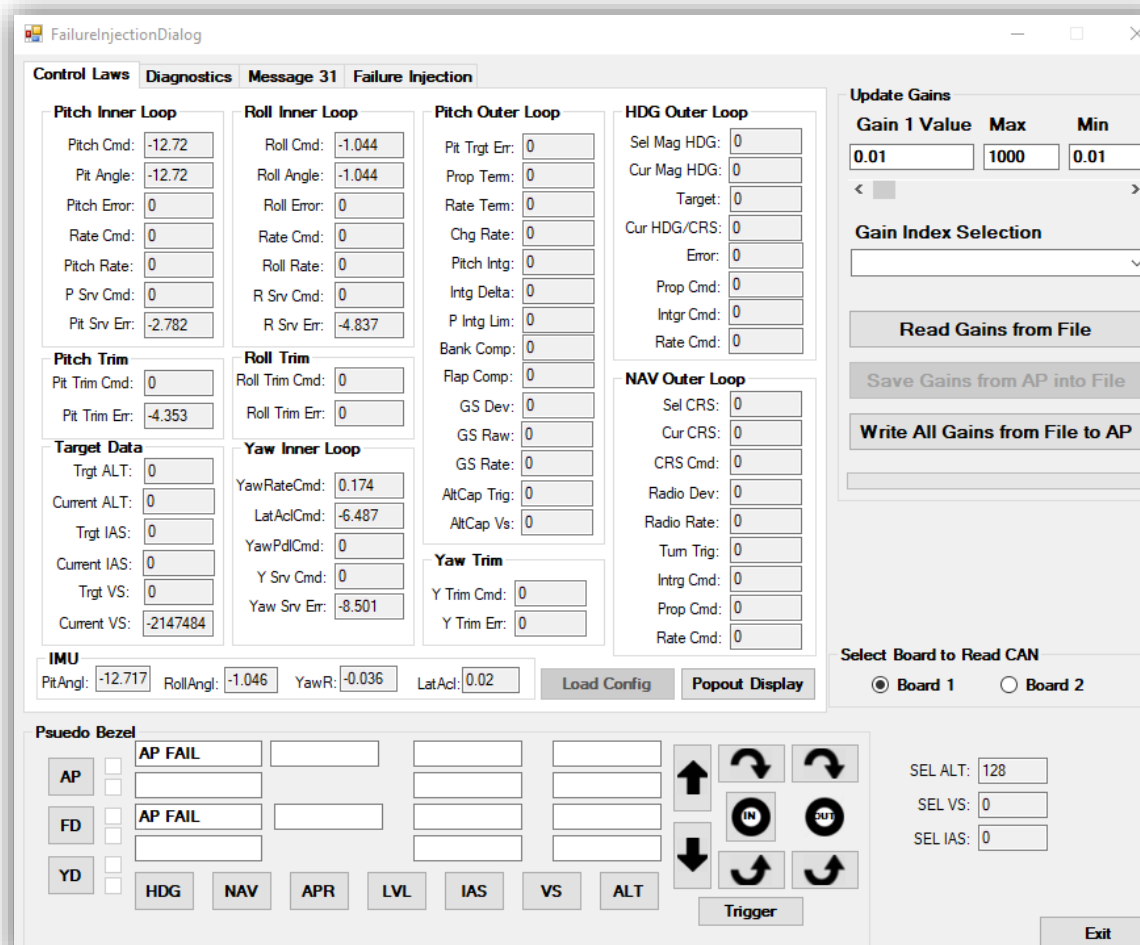
- Select the Ap5000CanConfigData.csv file
- Click Open



SOFTWARE TOOLS – TSOVISTA

TSOVista Introduction – Flight Test Tool

- At this point, many the data fields on the Control Laws tab should populate, and the Pseudo Bezel should display current modes matching the FGC display



The screenshot displays the 'FailureInjectionDialog' window with the 'Control Laws' tab selected. The interface is organized into several sections for configuring control laws and diagnostics.

Control Laws

- Pitch Inner Loop:** Pitch Cmd: -12.72, Pit Angle: -12.72, Pitch Error: 0, Rate Cmd: 0, Pitch Rate: 0, P Srv Cmd: 0, Pit Srv Err: -2.782.
- Roll Inner Loop:** Roll Cmd: -1.044, Roll Angle: -1.044, Roll Error: 0, Rate Cmd: 0, Roll Rate: 0, R Srv Cmd: 0, R Srv Err: -4.837.
- Pitch Outer Loop:** Pit Trgt Err: 0, Prop Term: 0, Rate Term: 0, Chg Rate: 0, Pitch Intg: 0, Intg Delta: 0, P Intg Lim: 0, Bank Comp: 0, Flap Comp: 0, GS Dev: 0, GS Raw: 0, GS Rate: 0, AltCap Trig: 0, AltCap Vs: 0.
- HDG Outer Loop:** Sel Mag HDG: 0, Cur Mag HDG: 0, Target: 0, Cur HDG/CRS: 0, Error: 0, Prop Cmd: 0, Intrg Cmd: 0, Rate Cmd: 0.
- NAV Outer Loop:** Sel CRS: 0, Cur CRS: 0, CRS Cmd: 0, Radio Dev: 0, Radio Rate: 0, Turn Trig: 0, Intrg Cmd: 0, Prop Cmd: 0, Rate Cmd: 0.
- Pitch Trim:** Pit Trim Cmd: 0, Pit Trim Err: -4.353.
- Roll Trim:** Roll Trim Cmd: 0, Roll Trim Err: 0.
- Yaw Inner Loop:** YawRateCmd: 0.174, LatAclCmd: -6.487, YawPdCmd: 0, Y Srv Cmd: 0, Yaw Srv Err: -8.501.
- Yaw Trim:** Y Trim Cmd: 0, Y Trim Err: 0.
- Target Data:** Trgt ALT: 0, Current ALT: 0, Trgt IAS: 0, Current IAS: 0, Trgt VS: 0, Current VS: -2147484.

IMU: PitAngl: -12.717, RollAngl: -1.046, YawR: -0.036, LatAcl: 0.02.

Update Gains: Gain 1 Value: 0.01, Max: 1000, Min: 0.01.

Gain Index Selection: (Dropdown menu)

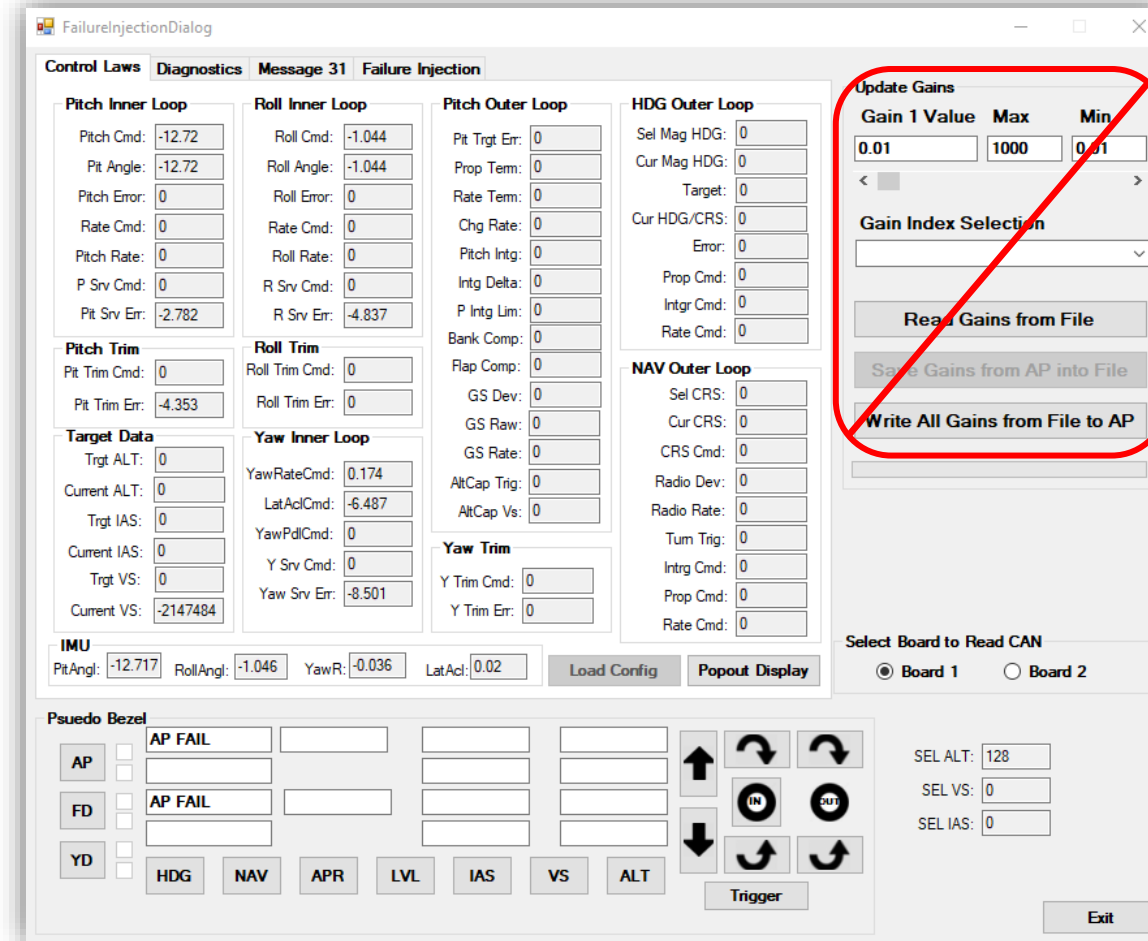
Buttons: Read Gains from File, Save Gains from AP into File, Write All Gains from File to AP.

Select Board to Read CAN: Board 1 (selected), Board 2.

Pseudo Bezel: AP, FD, YD, HDG, NAV, APR, LVL, IAS, VS, ALT, SEL ALT: 128, SEL VS: 0, SEL IAS: 0, Trigger, Exit.

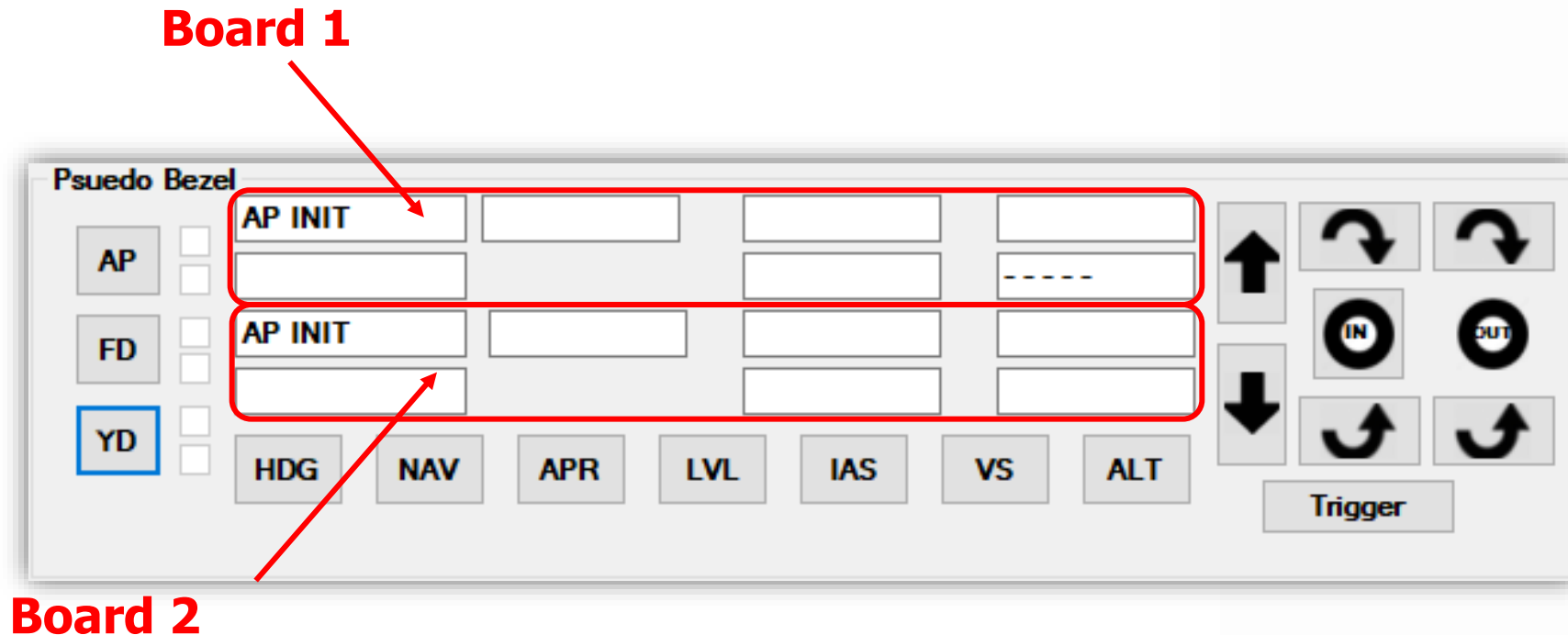
TSOVista Introduction – Flight Test Tool

- Do not attempt to manipulate the Update Gains section of the tool. This section is not active, and not authorized for field use.



TSOVista Introduction – Flight Test Tool

- The Pseudo Bezel Displays modes and targets for Board 1 and Board 2
 - A single board (no yaw) FGC will show AP FAIL for Board 2 at all times



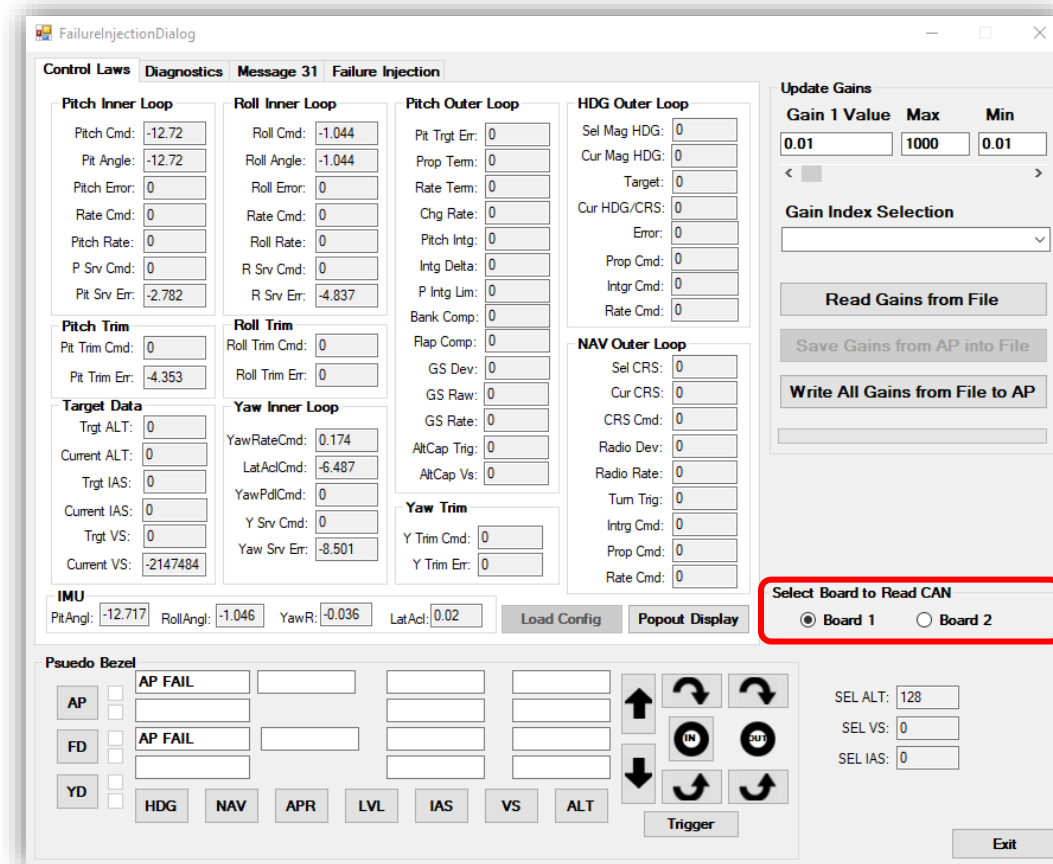
TSOVista Introduction – Flight Test Tool

- Clicking the 'Trigger' button disables the FGC bezel control, and enables FGC mode control via the Pseudo Bezel.
 - This is rarely, if ever, useful in the field, and should be avoided during flight.



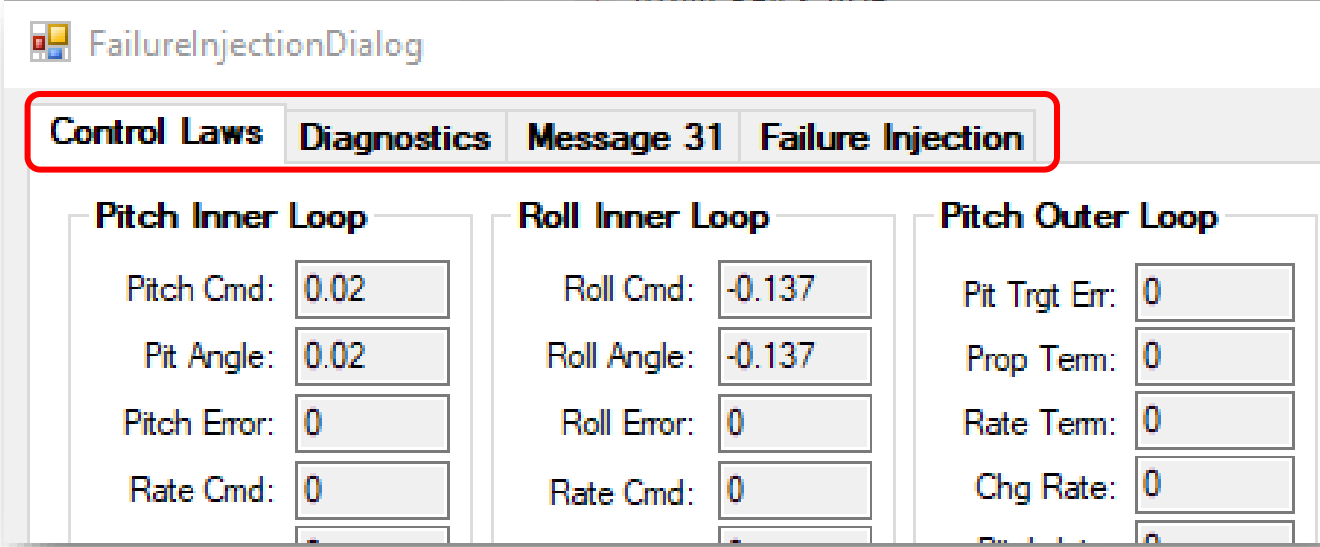
TSOVista Introduction – Flight Test Tool

- With the exception of the Pseudo Bezel, data displayed in this tool is read from the board selected in the 'Select Board to Read CAN' area.
- Further explanation of the Control Laws tab is beyond the scope of this presentation.



TSOVista Introduction – Flight Test Tool

- The tabs across the top of the window access additional information from the FGC via the CAN2 bus.

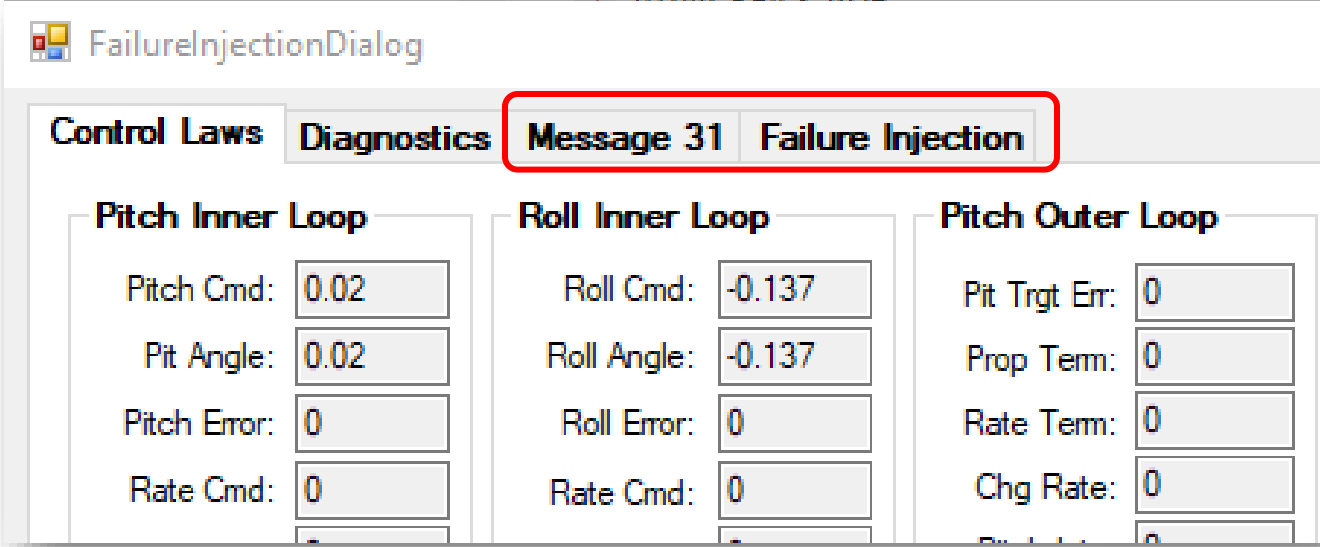


The screenshot shows a window titled "FailureInjectionDialog" with four tabs: "Control Laws", "Diagnostics", "Message 31", and "Failure Injection". The "Failure Injection" tab is selected and highlighted with a red box. Below the tabs, there are three columns of controls:

Pitch Inner Loop	Roll Inner Loop	Pitch Outer Loop
Pitch Cmd: 0.02	Roll Cmd: -0.137	Pit Trgt Err: 0
Pit Angle: 0.02	Roll Angle: -0.137	Prop Term: 0
Pitch Error: 0	Roll Error: 0	Rate Term: 0
Rate Cmd: 0	Rate Cmd: 0	Chg Rate: 0

TSOVista Introduction – Flight Test Tool

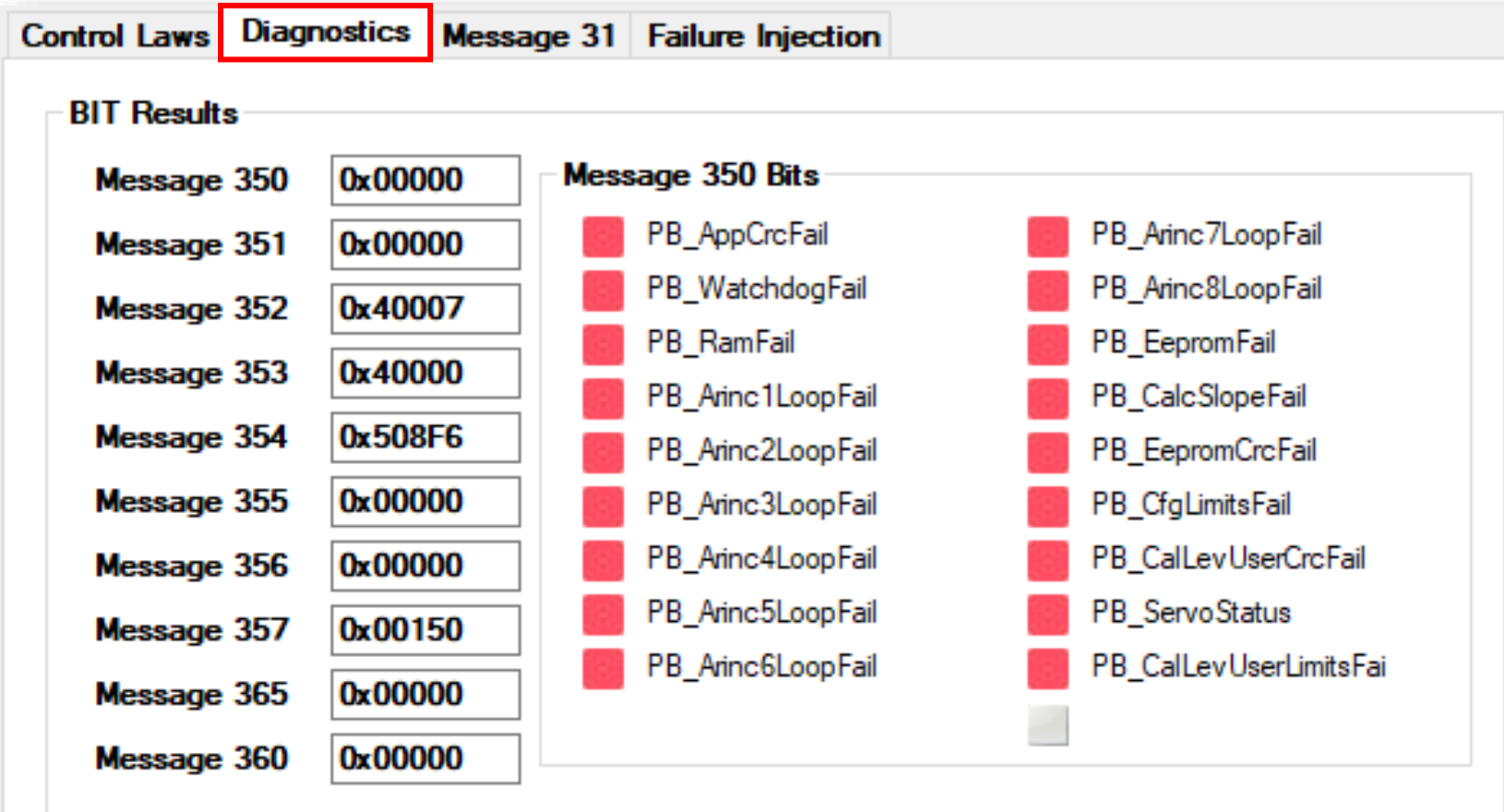
- The Message 31 and Failure Injection tabs are beyond the scope of this presentation.
- They are intended for development use by our engineering team, and don't have much if any value in the field.



Pitch Inner Loop		Roll Inner Loop		Pitch Outer Loop	
Pitch Cmd:	0.02	Roll Cmd:	-0.137	Pit Trgt Err:	0
Pit Angle:	0.02	Roll Angle:	-0.137	Prop Term:	0
Pitch Error:	0	Roll Error:	0	Rate Term:	0
Rate Cmd:	0	Rate Cmd:	0	Chg Rate:	0

TSOVista Introduction – Flight Test Tool

- Selecting the Diagnostics tab displays the BIT Results data.



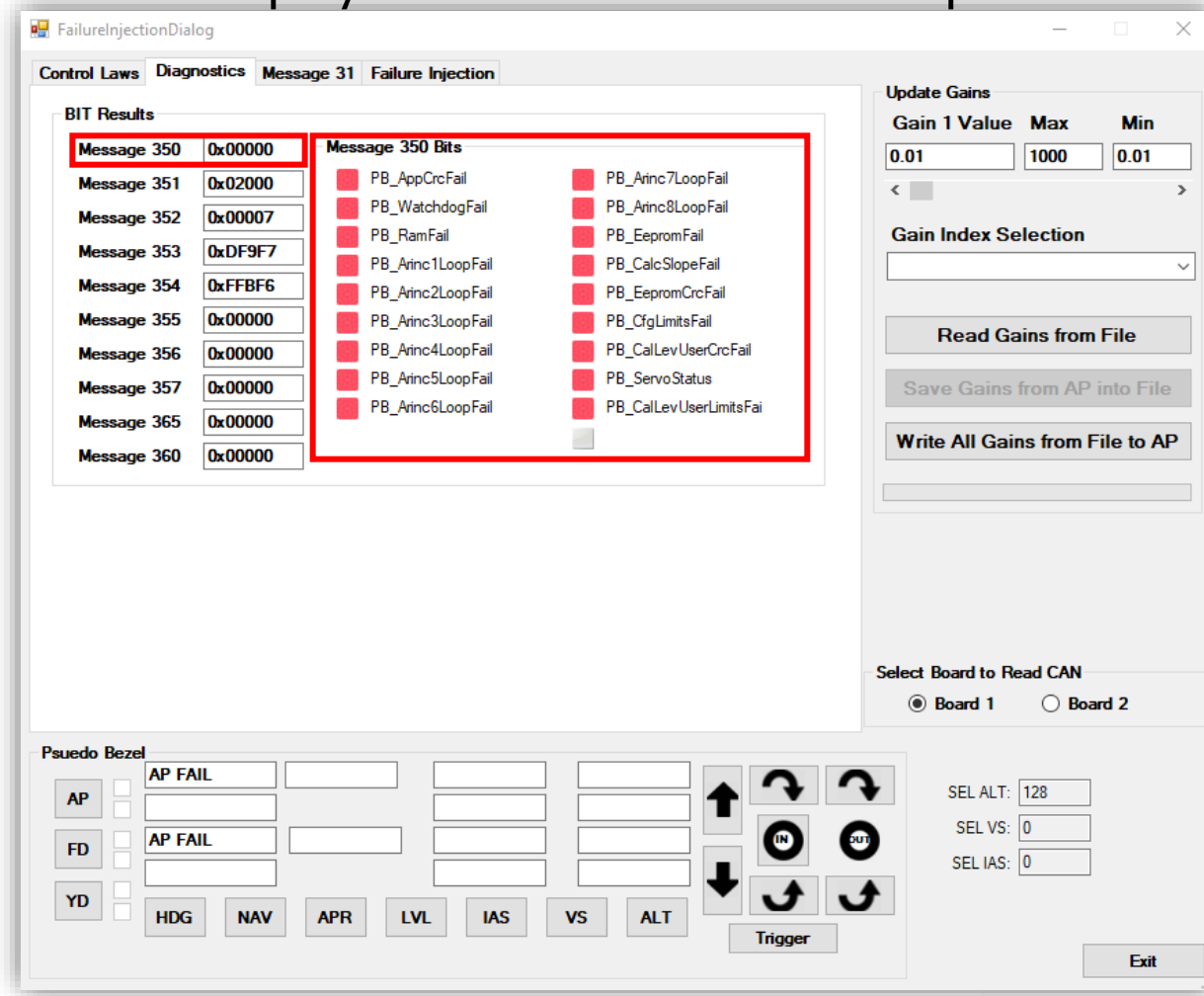
The screenshot shows the 'Diagnostics' tab selected in the TSOVista software. The 'BIT Results' section displays a list of messages and their corresponding bit patterns. The 'Message 350 Bits' section shows a list of failure codes, each preceded by a red square indicating a failed bit.

Message	BIT Results
Message 350	0x00000
Message 351	0x00000
Message 352	0x40007
Message 353	0x40000
Message 354	0x508F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00150
Message 365	0x00000
Message 360	0x00000

Message 350 Bits
☒ PB_AppCrcFail
☒ PB_WatchdogFail
☒ PB_RamFail
☒ PB_Arinc1LoopFail
☒ PB_Arinc2LoopFail
☒ PB_Arinc3LoopFail
☒ PB_Arinc4LoopFail
☒ PB_Arinc5LoopFail
☒ PB_Arinc6LoopFail
☒ PB_Arinc7LoopFail
☒ PB_Arinc8LoopFail
☒ PB_EepromFail
☒ PB_CalcSlopeFail
☒ PB_EepromCrcFail
☒ PB_CfgLimitsFail
☒ PB_CalLevUserCrcFail
☒ PB_ServoStatus
☒ PB_CalLevUserLimitsFai

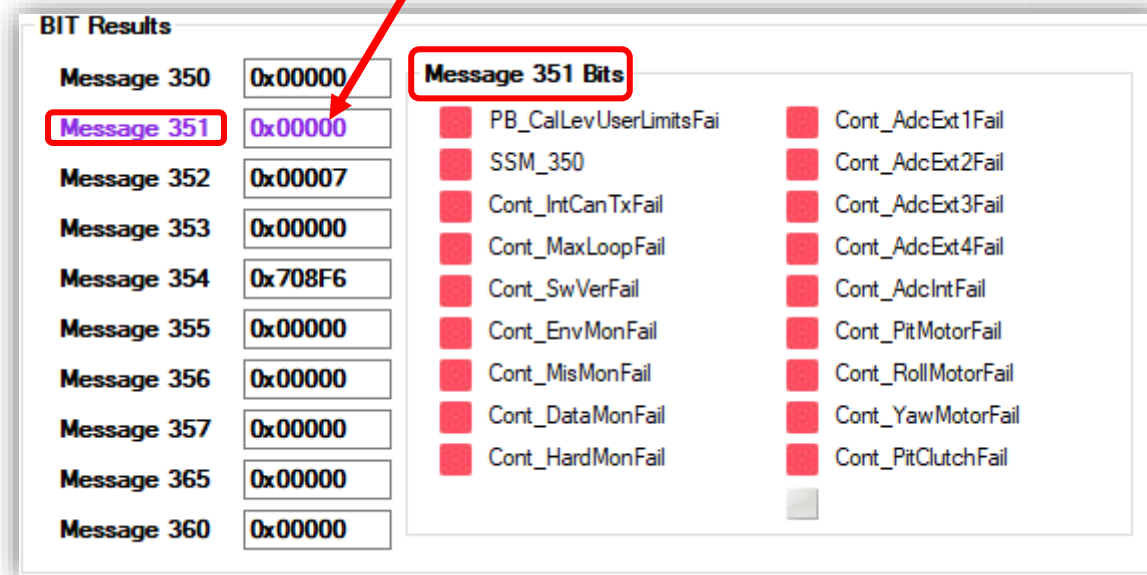
TSOVista Introduction – Flight Test Tool

- By default, Message 350 Bits are displayed when the tab is first opened



TSOVista Introduction – Flight Test Tool

- Selecting each Message will display the associated Message Bits
- The hexadecimal (HEX) value to the right of the Message 3** is a good indicator if a bit is flagged on that Message.
 - With one exception, a HEX value of 0x00000 indicates no failed Message Bits for that Message.



TSOVista Introduction – Flight Test Tool

- Selecting each Message will display the associated Message Bits
- The hexadecimal (HEX) value to the right of the Message 3** is a good indicator if a bit is flagged on that Message.
 - With one exception, a HEX value of 0x00000 indicates no failed Message Bits for that Message.
 - With SW V1.4(.1), a HEX value of 0X00007 for Message 352 indicates no failed message bits.

BIT Results	
Message 350	0x00000
Message 351	0x00000
Message 352	0x00007
Message 353	0x00000
Message 354	0x708F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 352 Bits	
☒ Cont_TrimFail	☐
☒ Cont_BezelFail	☐
☒ Cont_LowVoltageFail	☐
☒ Cont_AccelFail	☐
☒ Cont_5VRefFail	☐
☐	☐
☐	☐
☐	☐
☐	☐

TSOVista Introduction – Flight Test Tool

- Messages 353 and 354 can be ignored for diagnosing AP FAIL
 - These bits indicate the status of various ARINC 429 data labels
 - Not all ARINC 429 data labels are used in every configuration
 - Reference S-Tec document 87338 for ARINC 429 label requirements
 - Missing or invalid ARINC 429 data will not result in AP FAIL

BIT Results	
Message 350	0x00000
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 353 Bits	
☒ DM_VNV Altitude Target	☒ DM_SelHdgFail (101)
☒ DM_VNV Flight Path Ang	☒ DM_SelCourseFail (100)
☐ SSM_352	☐ DM_NomAccelFail (333)
☒ DM_VOR Omnibearing Fa	☐ DM_LatAccelFail (332)
☒ DM_Baro-Corrected Alti	☒ DM_YawRateFail (330)
☒ DM_Glideslope Deviatio	☒ DM_RollRateFail (327)
☒ DM_Lateral Deviation F	☒ DM_PitchRateFail (326)
☒ DM_Radar Altitude Fail	☒ DM_RollAngFail (325)
☒ DM_HorCommSigFail (121)	☒ DM_PitchAngFail (324)
	☒ DM_MagHdgFail (320)

TSOVista Introduction – Flight Test Tool

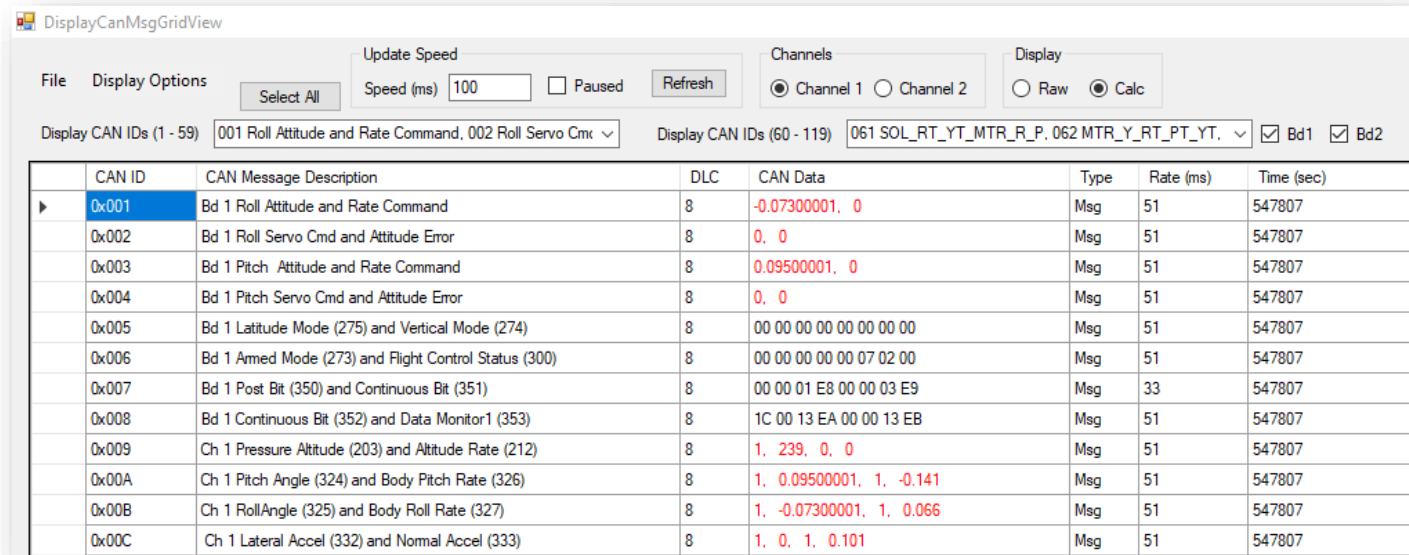
- Messages 353 and 354 can be ignored for diagnosing AP FAIL
 - These bits indicate the status of various ARINC 429 data labels
 - Not all ARINC 429 data labels are used in every configuration
 - Reference S-Tec document 87338 for ARINC 429 label requirements
 - Missing or invalid ARINC 429 data will not result in AP FAIL

BIT Results	
Message 350	0x00000
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 354 Bits	
☒ DM_TRUE_HDG (314)	☒ DM_Ground Speed Fail (
☒ DM_GS_ANG (155)	☒ DM_True Track Angle Fa
☐ DM_TAS (210)	☒ DM_FMS Config Fail (26
☒ DM_HDG_DTM (105)	☒ DM_Magnetic Variation
☒ DM_COURSE_DTM (277)	☒ DM_VNAV Deviation Fail
☒ DM_Sel VS (104)	☒ DM_LNAV Deviation Fail
☒ DM_Sel IAS (103)	☒ DM_Desired Track Angle
☒ DM_Sel Alt (102)	☐
☐ DM_Vert Prop Dev Fail	☐
	☐

TSOVista Introduction – CanGridViewDisplay

- View CAN messages in real time including:
 - state of discrete inputs
 - IMU data
 - ARINC 429 input data
 - Analog inputs (HDG/CRS error, CDI/GDI, etc)
 - Servo motor and solenoid current and voltages
 - Etc..
- Record CAN messages for engineering evaluation

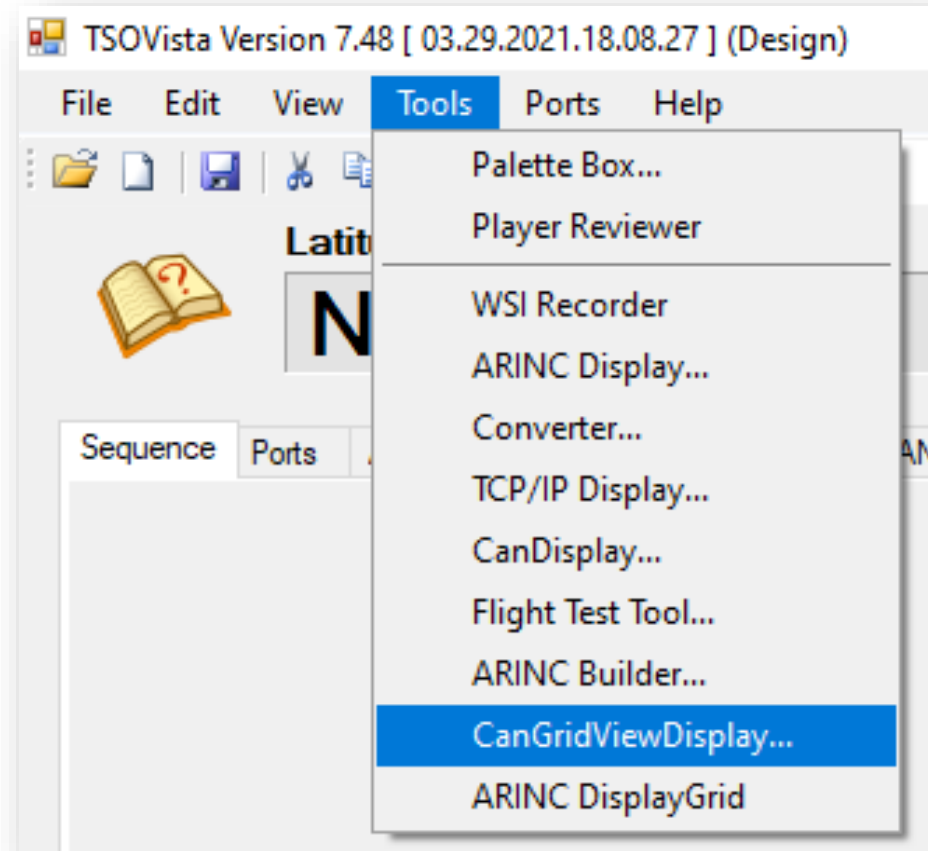


The screenshot shows the 'DisplayCanMsgGridView' application window. It features a menu bar with 'File' and 'Display Options'. Below the menu bar, there are controls for 'Update Speed' (Speed (ms) 100, Paused checkbox, Refresh button), 'Channels' (Channel 1 selected, Channel 2 unselected), and 'Display' (Raw unselected, Calc selected). The main display area shows two dropdown menus for 'Display CAN IDs (1 - 59)' and 'Display CAN IDs (60 - 119)', both with 'Select All' buttons. Below these are checkboxes for 'Bd1' and 'Bd2'. The main data table has columns: CAN ID, CAN Message Description, DLC, CAN Data, Type, Rate (ms), and Time (sec). The table lists 12 messages, with the first message (0x001) highlighted in blue.

	CAN ID	CAN Message Description	DLC	CAN Data	Type	Rate (ms)	Time (sec)
▶	0x001	Bd 1 Roll Attitude and Rate Command	8	-0.07300001, 0	Msg	51	547807
	0x002	Bd 1 Roll Servo Cmd and Attitude Error	8	0, 0	Msg	51	547807
	0x003	Bd 1 Pitch Attitude and Rate Command	8	0.09500001, 0	Msg	51	547807
	0x004	Bd 1 Pitch Servo Cmd and Attitude Error	8	0, 0	Msg	51	547807
	0x005	Bd 1 Latitude Mode (275) and Vertical Mode (274)	8	00 00 00 00 00 00 00 00	Msg	51	547807
	0x006	Bd 1 Amed Mode (273) and Flight Control Status (300)	8	00 00 00 00 00 07 02 00	Msg	51	547807
	0x007	Bd 1 Post Bit (350) and Continuous Bit (351)	8	00 00 01 E8 00 00 03 E9	Msg	33	547807
	0x008	Bd 1 Continuous Bit (352) and Data Monitor1 (353)	8	1C 00 13 EA 00 00 13 EB	Msg	51	547807
	0x009	Ch 1 Pressure Altitude (203) and Altitude Rate (212)	8	1, 239, 0, 0	Msg	51	547807
	0x00A	Ch 1 Pitch Angle (324) and Body Pitch Rate (326)	8	1, 0.09500001, 1, -0.141	Msg	51	547807
	0x00B	Ch 1 RollAngle (325) and Body Roll Rate (327)	8	1, -0.07300001, 1, 0.066	Msg	51	547807
	0x00C	Ch 1 Lateral Accel (332) and Normal Accel (333)	8	1, 0, 1, 0.101	Msg	51	547807

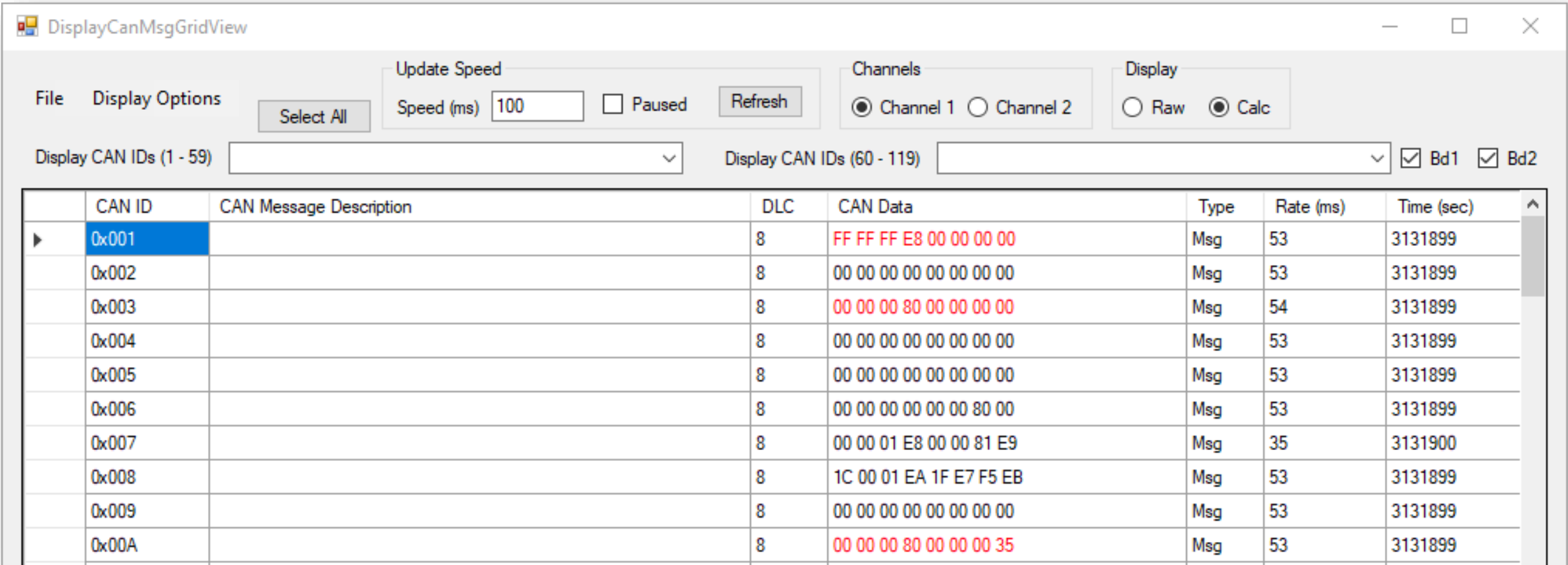
TSOVista Introduction – CanGridViewDisplay

- From the Menu bar, select Tools>CanGridViewDisplay...



TSOVista Introduction – CanGridViewDisplay

- Data should now be visible, but is not useable.

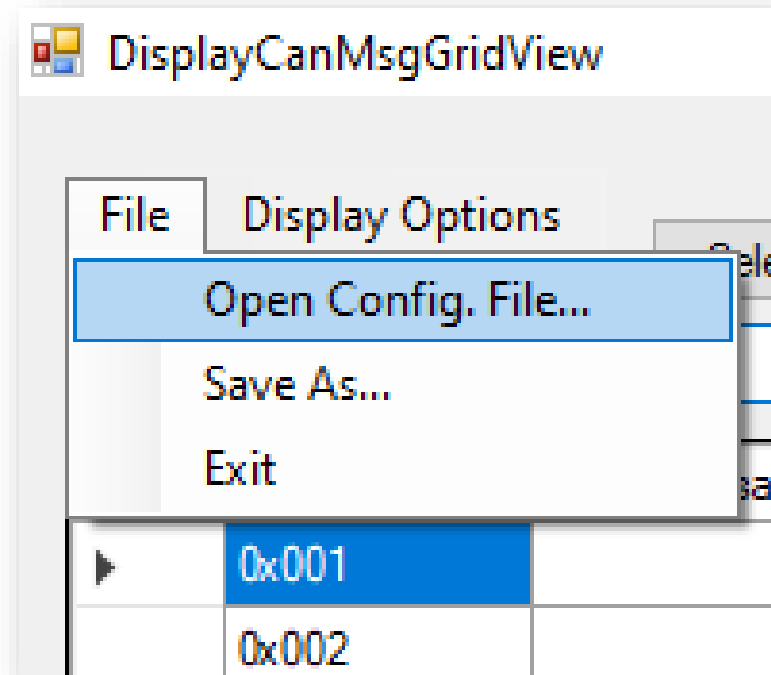


The screenshot shows the 'DisplayCanMsgGridView' application window. The interface includes a menu bar with 'File' and 'Display Options'. Below the menu bar, there are several control groups: 'Update Speed' with a 'Speed (ms)' input set to 100, a 'Paused' checkbox, and a 'Refresh' button; 'Channels' with radio buttons for 'Channel 1' (selected) and 'Channel 2'; and 'Display' with radio buttons for 'Raw' and 'Calc' (selected). There are also two dropdown menus for 'Display CAN IDs (1 - 59)' and 'Display CAN IDs (60 - 119)', and checkboxes for 'Bd1' and 'Bd2'. The main area is a table with the following columns: CAN ID, CAN Message Description, DLC, CAN Data, Type, Rate (ms), and Time (sec). The table contains 11 rows of data, with the first row (0x001) highlighted in blue. The CAN Data for the first row is 'FF FF FF E8 00 00 00 00' in red text.

	CAN ID	CAN Message Description	DLC	CAN Data	Type	Rate (ms)	Time (sec)
▶	0x001		8	FF FF FF E8 00 00 00 00	Msg	53	3131899
	0x002		8	00 00 00 00 00 00 00 00	Msg	53	3131899
	0x003		8	00 00 00 80 00 00 00 00	Msg	54	3131899
	0x004		8	00 00 00 00 00 00 00 00	Msg	53	3131899
	0x005		8	00 00 00 00 00 00 00 00	Msg	53	3131899
	0x006		8	00 00 00 00 00 00 00 80	Msg	53	3131899
	0x007		8	00 00 01 E8 00 00 81 E9	Msg	35	3131900
	0x008		8	1C 00 01 EA 1F E7 F5 EB	Msg	53	3131899
	0x009		8	00 00 00 00 00 00 00 00	Msg	53	3131899
	0x00A		8	00 00 00 80 00 00 00 35	Msg	53	3131899

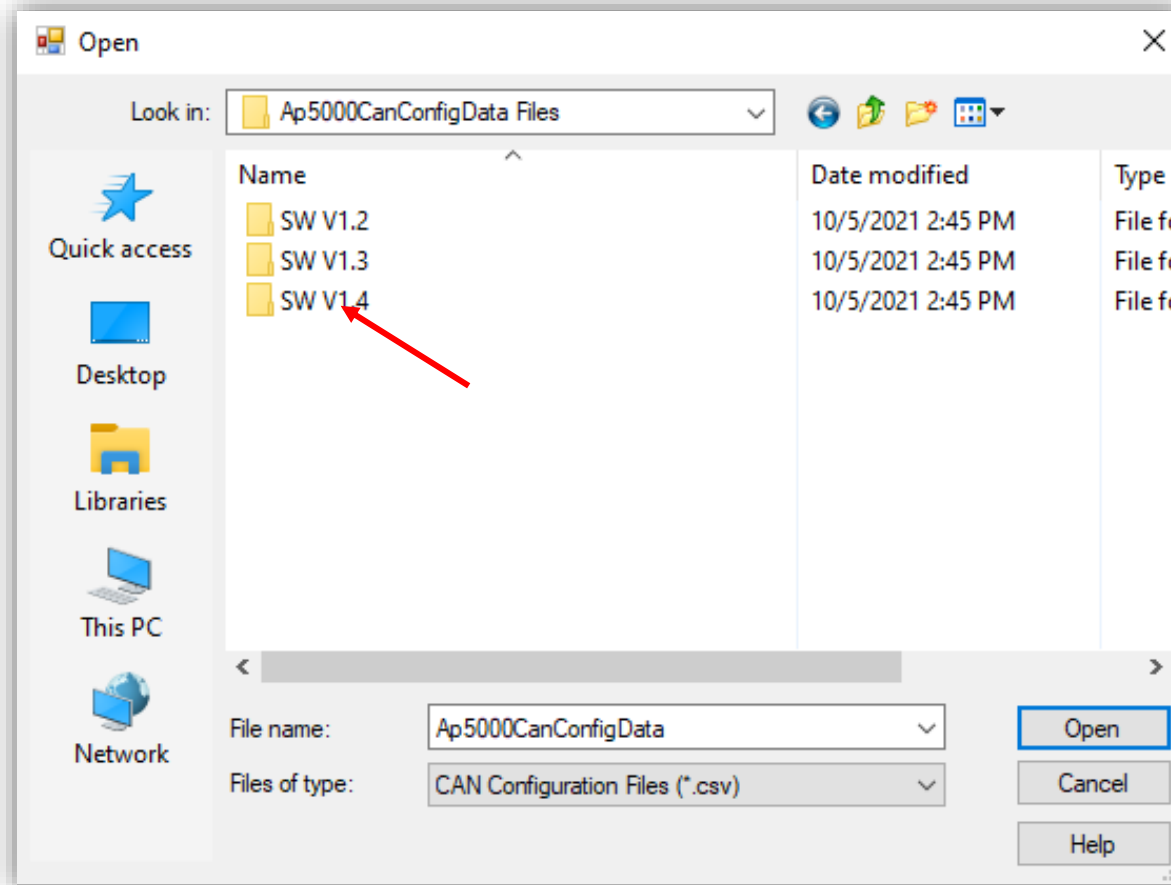
TSOVista Introduction – CanGridViewDisplay

- Select File> Open Config.File...



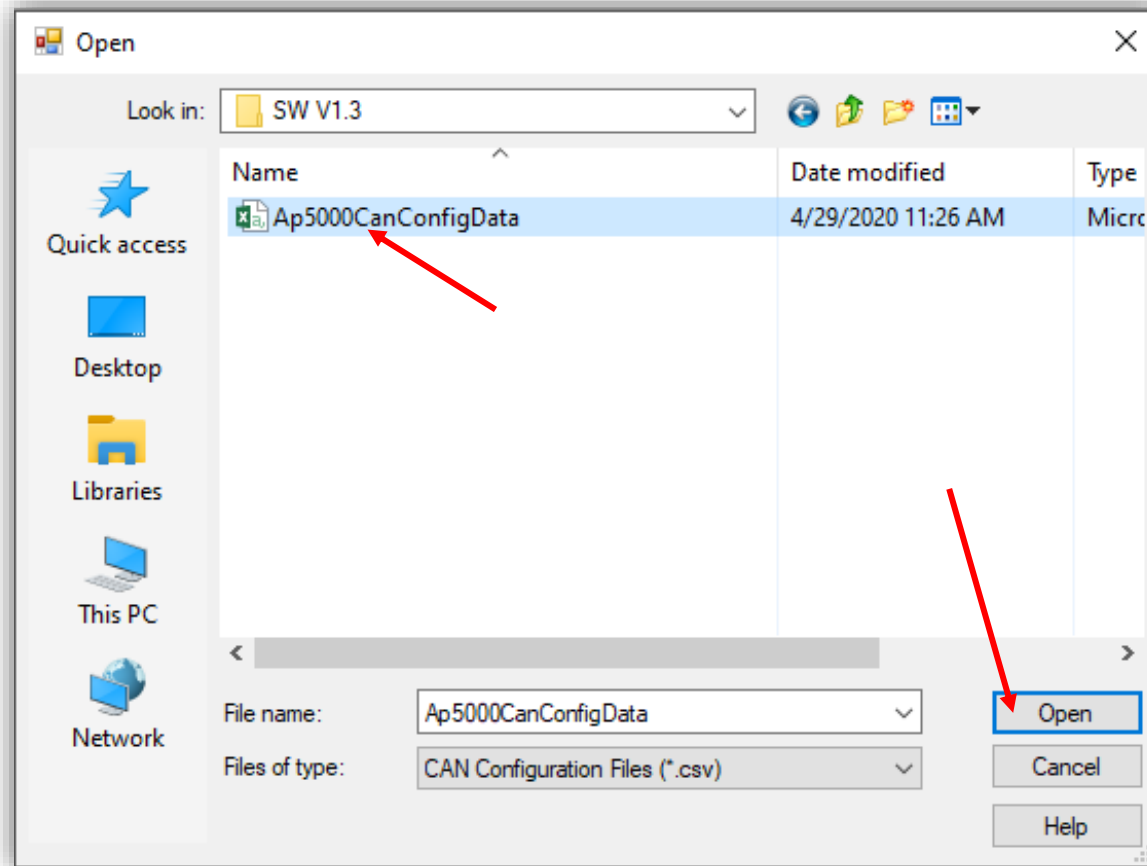
TSOVista Introduction – CanGridViewDisplay

- Open the SW 1.X folder that corresponds to the Application SW version on the FGC



TSOVista Introduction – CanGridViewDisplay

- Select the Ap5000CanConfigData.csv file
- Click Open



TSOVista Introduction – CanGridViewDisplay

- CAN data should now be visible and readable
- Feel free to explore

DisplayCanMsgGridView

File Display Options Update Speed Speed (ms) ☐ Paused Channels ☒ Channel 1 ☐ Channel 2 Display ☐ Raw ☒ Calc

Display CAN IDs (1 - 59) Display CAN IDs (60 - 119) ☒ Bd1 ☒ Bd2

	CAN ID	CAN Message Description	DLC	CAN Data	Type	Rate (ms)	Time (sec)
▶	0x001	Bd 1 Roll Attitude and Rate Command	8	-0.07300001, 0	Msg	51	547807
	0x002	Bd 1 Roll Servo Cmd and Attitude Error	8	0, 0	Msg	51	547807
	0x003	Bd 1 Pitch Attitude and Rate Command	8	0.09500001, 0	Msg	51	547807
	0x004	Bd 1 Pitch Servo Cmd and Attitude Error	8	0, 0	Msg	51	547807
	0x005	Bd 1 Latitude Mode (275) and Vertical Mode (274)	8	00 00 00 00 00 00 00 00	Msg	51	547807
	0x006	Bd 1 Armed Mode (273) and Flight Control Status (300)	8	00 00 00 00 00 07 02 00	Msg	51	547807
	0x007	Bd 1 Post Bit (350) and Continuous Bit (351)	8	00 00 01 E8 00 00 03 E9	Msg	33	547807
	0x008	Bd 1 Continuous Bit (352) and Data Monitor1 (353)	8	1C 00 13 EA 00 00 13 EB	Msg	51	547807
	0x009	Ch 1 Pressure Altitude (203) and Altitude Rate (212)	8	1, 239, 0, 0	Msg	51	547807
	0x00A	Ch 1 Pitch Angle (324) and Body Pitch Rate (326)	8	1, 0.09500001, 1, -0.141	Msg	51	547807
	0x00B	Ch 1 RollAngle (325) and Body Roll Rate (327)	8	1, -0.07300001, 1, 0.066	Msg	51	547807
	0x00C	Ch 1 Lateral Accel (332) and Normal Accel (333)	8	1, 0, 1, 0.101	Msg	51	547807

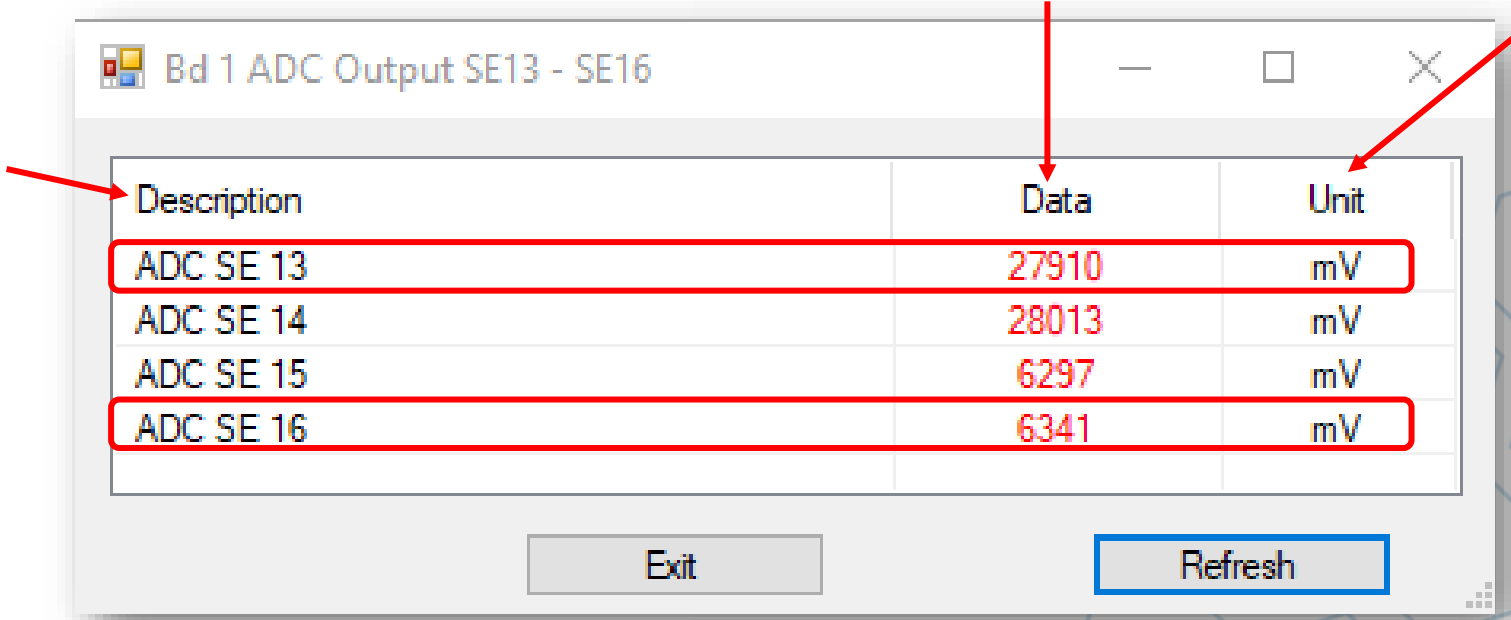
TSOVista Introduction – CanGridViewDisplay

- Some useful data:
- ADC SE1 - SE31
 - Analog to Digital Converter Single Ended
 - These are discreet inputs
 - Double Click on a line will pop a smaller window with additional data

CAN ID	CAN Message Description	DLC	CAN Data
0x030	Bd 1 ADC Output SE1 - SE4	8	14, 6312, 6360, 6312
0x031	Bd 1 ADC Output SE5 - SE8	8	6312, 6304, 6339, 6311
0x032	Bd 1 ADC Output SE9 - SE12	8	6334, 6341, 6325, 6343
0x033	Bd 1 ADC Output SE13 - SE16	8	27910, 28020, 6290, 6334
0x034	Bd 1 ADC Output SE17 - SE20	8	6304, 6326, 6319, 6348
0x035	Bd 1 ADC Output SE21 - SE24	8	6326, 6326, 6304, 6348
0x036	Bd 1 ADC Output SE25 - SE28	8	6326, 2140, 29, 29
0x037	Bd 1 ADC Output SE29 - SE31	8	6334, 29, 6246

TSOVista Introduction – CanGridViewDisplay

- Some useful data:
 - Description column is the discreet name
 - Data column contains the real time value
 - Unit provides the unit of measure
 - ADC SE 13 has a value of 27,910 mVDC, or aprox 28 VDC
 - This input has 28 VDC applied to it, or HIGH
 - ADC SE 16 has a value of 6341, or aprox 6.3 VDC
 - This is a floating value, meaning it is open, not grounded or powered.

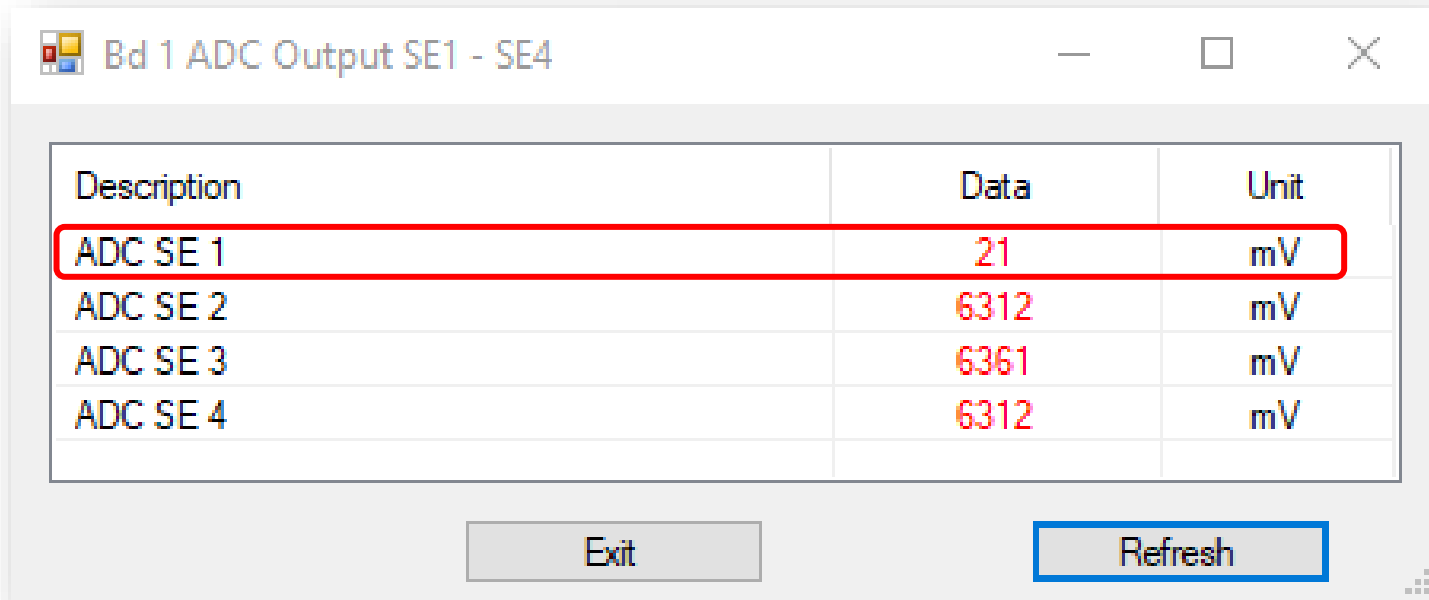


Description	Data	Unit
ADC SE 13	27910	mV
ADC SE 14	28013	mV
ADC SE 15	6297	mV
ADC SE 16	6341	mV

Exit Refresh

TSOVista Introduction – CanGridViewDisplay

- Some useful data:
 - ADC SE 1 has a value of 21 mVDC, or 0.021 VDC
 - This discreet input is grounded, or LOW



The screenshot shows a software window titled "Bd 1 ADC Output SE1 - SE4". It contains a table with three columns: "Description", "Data", and "Unit". The first row, "ADC SE 1", is highlighted with a red border and shows a value of "21" in red text. The other rows show values of "6312" in red text. At the bottom of the window, there are two buttons: "Exit" and "Refresh".

Description	Data	Unit
ADC SE 1	21	mV
ADC SE 2	6312	mV
ADC SE 3	6361	mV
ADC SE 4	6312	mV

TSOVista Introduction – CanGridViewDisplay

- List of SE inputs:

SE Number	Pin	Mapped Signal	SE Number	Pin	Mapped Signal
SE 1	P1-11	Ice Protect In (FIKI)	SE 17	P1-39	Go Around Switch
SE 2	P1-32	Not Used	SE 18	P1-53	Not Used
SE 3	P1-33	Not Used	SE 19	P1-54	AHRS Select
SE 4	P1-34	Not Used	SE 20	P1-55	ADC Select
SE 5	P1-16	Trim Sense Switch 1	SE 21	P1-57	ST-360 Enable (ST-360 only)
SE 6	P1-17	Trim Sense Switch 2	SE 22	P1-58	EFIS Select
SE 7	P1-18	Manual Elec Trim 1	SE 23	P1-59	Approach Select 2
SE 8	P1-19	Manual Elec Trim 2	SE 24	P1-69	GPS Mode Select 2
SE 9	P1-20	Not Used	SE 25	P1-70	HSI Valid
SE 10	P1-32	Not Used	SE 26	P1-71	VS Command (ST-360 only)
SE 11	P1-33	Not Used	SE 27	P1-72	Flap Pot In
SE 12	P1-34	Not Used	SE 28	P1-75	14 VDC Enable
SE 13	P1-35	AP Disconnect	SE 29	P1-74	GPS Mode Select 1
SE 14	P1-36	Trim Interrupt	SE 30	P1-76	Half Bank In
SE 15	P1-37	Approach Select 1	SE 31	P1-77	ALT Engage (ST-360 only)
SE 16	P1-38	CWS			

TSOVista Introduction – CanGridViewDisplay

- Additional useful data:
 - Both CAN ID 0x058 and 0x158 display board 1 and board 2 power (P1-1, P2-78)
 - In this case, Board 1 (P1-1) has 28 VDC applied

CAN ID	CAN Message Description	DLC	CAN Data
0x058	Bd 1 28V Power (Dbg_37)	8	28032.35, 1638.731
0x059	Bd 1 5V Refer		
0x05A	Bd 1 External		
0x05B	Bd 1 External		
0x05C	Bd 1 Roll and		
0x05D	Bd 1 Yaw and		
0x05E	Bd 1 Yaw Trim		
0x05F	Bd 1 Roll and		
0x060	Bd 1 Yaw and		
0x061	Bd 1 Yaw Trim		

Description	Data	Unit
Board 1 28V Power (Dbg37_0)	28032.35	value
Board 2 28V Power (Dbg37_1)	1638.731	value

Exit Refresh

TSOVista Introduction – CanGridViewDisplay

- Additional useful data:
 - Both CAN ID 0x058 and 0x158 display board 1 and board 2 power (P1-1, P2-78)
 - Board 2 (P2-78) is not powered (aprox 1.6 VDC)

CAN ID	CAN Message Description	DLC	CAN Data
0x058	Bd 1 28V Power (Dbg_37)	8	28032.35, 1638.731
0x059	Bd 1 5V Refer		
0x05A	Bd 1 External		
0x05B	Bd 1 External		
0x05C	Bd 1 Roll and		
0x05D	Bd 1 Yaw and		
0x05E	Bd 1 Yaw Trim		
0x05F	Bd 1 Roll and		
0x060	Bd 1 Yaw and		
0x061	Bd 1 Yaw Trim		


Bd 1 28V Power (Dbg_37)
—
□
✕

Description	Data	Unit
Board 1 28V Power (Dbg37_0)	28032.35	value
Board 2 28V Power (Dbg37_1)	1638.731	value

Exit
Refresh



Now we can use TSOVista to help troubleshoot!

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 350 $\neq$ 0x00000:

✓ PB_ServoStatus

Look for other failed messages

✓ Message 351 $\neq$ 0x00000

✓ Message 355 $\neq$ 0x00000



BIT Results

Message	Value
Message 350	0x00001
Message 351	0x02020
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00810
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 350 Bits

☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 351 $\neq$ 0x00000
 - ✓ Cont_DataMonFail (present with most AP FAILs)
 - ✓ Cont_RollMotorFail



BIT Results

Message	Value
Message 350	0x00001
Message 351	0x02020
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00810
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits

☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☒ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 355 $\neq$ 0x00000
 - ✓ DM_PFD Configuration
 - ✓ Roll Axis Motor Cmd
 - ✓ Roll Axis POST Motor



BIT Results

Message	Value
Message 350	0x00001
Message 351	0x02020
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00810
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 355 Bits

Bit	Status	Description
DM_PFD Configuration F	✓	Pitch Axis Clutch Enga
SSM_354		Roll Axis Insufficient
Pitch Axis Insufficien		Roll Axis Solenoid Ove
Pitch Axis Solenoid Ov		Roll Axis Solenoid Und
Pitch Axis Solenoid Un		Roll Axis Motor Cmd (V
Pitch Axis Motor Cmd (	✓	Roll Axis POST Motor C
Pitch Axis POST Motor		Roll Axis POST Motor V
Pitch Axis POST Motor		
Pitch Axis POST Motor		

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_RollMotorFail
- ✓ Message 355 - Roll Axis Motor Cmd
- ✓ Message 355 - Roll Axis POST Motor



Message 350 Bits

☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai
☐	☐

Message 351 Bits

☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☒ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail
☐	☐

Message 355 Bits

☒ DM_PFD Configuration F	☐ Pitch Axis Clutch Enga
☐ SSM_354	☐ Roll Axis Insufficient
☐ Pitch Axis Insufficien	☐ Roll Axis Solenoid Ove
☐ Pitch Axis Solenoid Ov	☐ Roll Axis Solenoid Und
☐ Pitch Axis Solenoid Un	☒ Roll Axis Motor Cmd (V
☐ Pitch Axis Motor Cmd (	☒ Roll Axis POST Motor C
☐ Pitch Axis POST Motor	☐ Roll Axis POST Motor V
☐ Pitch Axis POST Motor	☐
☐	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Failed Bits

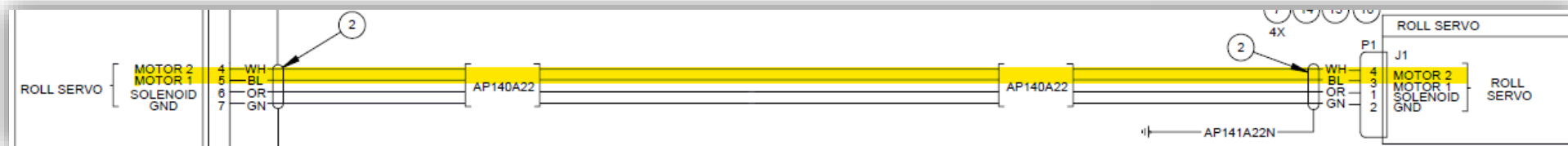
- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_RollMotorFail
- ✓ Message 355 - Roll Axis Motor Cmd
- ✓ Message 355 - Roll Axis POST Motor



Indicates the Roll servo motor failed the Power On Self Test (POST)

Corrective Action

- Check for failed Roll servo motor
 - Short/open/high resistance between servo pins 3 and 4
- Check wiring between FGC and Roll Servo
 - Shorts, opens, miss-wire
- Possible failed FGC



TROUBLESHOOTING AP FAIL (POST)



Congratulations!

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 350 $\neq$ 0x00000:

✓ PB_ServoStatus

Look for other failed messages

✓ Message 351 $\neq$ 0x00000

✓ Message 355 $\neq$ 0x00000



BIT Results			
Message 350	0x00001	Message 350 Bits	
Message 351	0x02010	☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
Message 352	0x00007	☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
Message 353	0xDF9F7	☐ PB_RamFail	☐ PB_EepromFail
Message 354	0xFFBF6	☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
Message 355	0x81000	☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
Message 356	0x00000	☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
Message 357	0x00000	☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
Message 365	0x00000	☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
Message 360	0x00000	☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 351 ≠ 0x00000
 - ✓ Cont_DataMonFail
 - ✓ Cont_PitMotorFail



BIT Results	
Message 350	0x00001
Message 351	0x02010
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x81000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☒ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 355 $\neq$ 0x00000
 - ✓ DM_PFD Configuration
 - ✓ Pitch Axis Motor Cmd
 - ✓ Pitch Axis POST Motor



BIT Results

Message	Value
Message 350	0x00001
Message 351	0x02010
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x81000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 355 Bits

Bit	Value	Description
✓	DM_PFD Configuration F	Pitch Axis Clutch Enga
✓	SSM_354	Roll Axis Insufficient
✓	Pitch Axis Insufficien	Roll Axis Solenoid Ove
✓	Pitch Axis Solenoid Ov	Roll Axis Solenoid Und
✓	Pitch Axis Solenoid Un	Roll Axis Motor Cmd (V
✓	Pitch Axis Motor Cmd (	Roll Axis POST Motor C
✓	Pitch Axis POST Motor	Roll Axis POST Motor V
✓	Pitch Axis POST Motor	
✓	Pitch Axis POST Motor	

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_PitMotorFail
- ✓ Message 355 - Pitch Axis Motor Cmd
- ✓ Message 355 - Pitch Axis POST Motor



Message 350 Bits

☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai
☐	☐

Message 351 Bits

☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☒ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail
☐	☐

Message 355 Bits

☒ DM_PFD Configuration F	☐ Pitch Axis Clutch Enga
☐ SSM_354	☐ Roll Axis Insufficient
☐ Pitch Axis Insufficien	☐ Roll Axis Solenoid Ove
☐ Pitch Axis Solenoid Ov	☐ Roll Axis Solenoid Und
☐ Pitch Axis Solenoid Un	☐ Roll Axis Motor Cmd (V
☒ Pitch Axis Motor Cmd (	☐ Roll Axis POST Motor C
☒ Pitch Axis POST Motor	☐ Roll Axis POST Motor V
☐ Pitch Axis POST Motor	☐
☐ Pitch Axis POST Motor	☐
☐	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

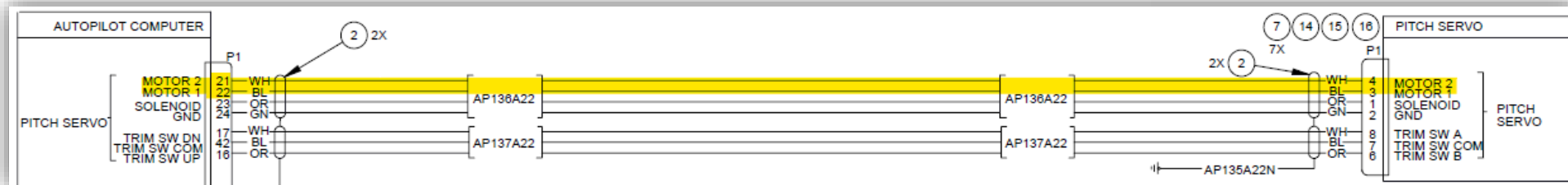
- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_PitMotorFail
- ✓ Message 355 - Pitch Axis Motor Cmd
- ✓ Message 355 - Pitch Axis POST Motor



Indicates the Pitch servo motor failed the Power On Self Test (POST)

Corrective Action

- Check for failed Pitch servo motor
 - Short/open/high resistance between servo pins 3 and 4
- Check wiring between FGC and Pitch Servo
 - Shorts, opens, miswire
- Possible failed FGC



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 350 $\neq$ 0x00000:

✓ PB_ServoStatus

Look for other failed messages

✓ Message 351 $\neq$ 0x00000

✓ Message 356 $\neq$ 0x00000



BIT Results	
Message 350	0x00001
Message 351	0x02040
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00810
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 350 Bits	
☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 351 $\neq$ 0x00000
 - ✓ Cont_DataMonFail
 - ✓ Cont_YawMotorFail



BIT Results	
Message 350	0x00001
Message 351	0x02040
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00810
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☒ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 356 ≠ 0x00000
 - ✓ Yaw Axis Motor Cmd
 - ✓ Yaw Axis POST Motor



BIT Results

Message 350	0x00001
Message 351	0x02040
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00810
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 356 Bits

☐	Roll Trim Axis Insuffi	☐	Yaw Axis Solenoid Over
☐	Roll Trim Axis Solenoi	☐	Yaw Axis Solenoid Unde
☐	Roll Trim Axis Solenoi	☒	Yaw Axis Motor Cmd (Vo
☐	Roll Trim Axis Motor C	☒	Yaw Axis POST Motor Cu
☐	Roll Trim Axis POST Mo	☐	Yaw Axis POST Motor Vo
☐	Roll Trim Axis POST Mo	☐	Yaw Axis POST Motor Vo
☐	Roll Trim Axis POST Mo	☐	Yaw Axis Clutch Engage
☐	Roll Trim Axis Clutch	☐	
☐	Yaw Axis Insufficient	☐	

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_YawMotorFail
- ✓ Message 356 - Yaw Axis Motor Cmd
- ✓ Message 356 - Yaw Axis POST Motor



Message 350 Bits

☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai

Message 351 Bits

☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☒ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Message 356 Bits

☐ Roll Trim Axis Insuffi	☐ Yaw Axis Solenoid Over
☐ Roll Trim Axis Solenoi	☐ Yaw Axis Solenoid Unde
☐ Roll Trim Axis Solenoi	☒ Yaw Axis Motor Cmd (Vo
☐ Roll Trim Axis Motor C	☒ Yaw Axis POST Motor Cu
☐ Roll Trim Axis POST Mo	☐ Yaw Axis POST Motor Vo
☐ Roll Trim Axis POST Mo	☐ Yaw Axis POST Motor Vo
☐ Roll Trim Axis POST Mo	☐ Yaw Axis Clutch Engage
☐ Roll Trim Axis Clutch	☐
☐ Yaw Axis Insufficient	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Failed Bits

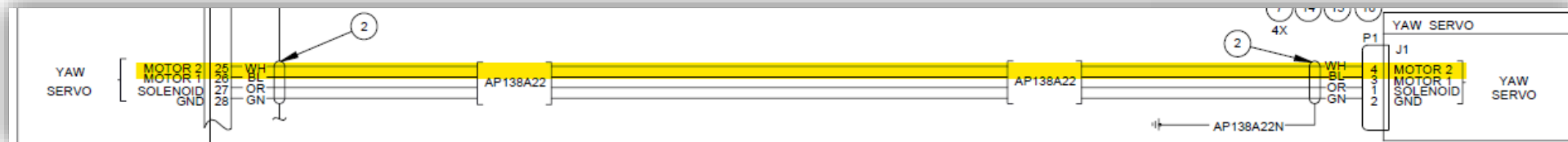
- ✓ Message 350 - PB_ServoStatus
- ✓ Message 351 - Cont_YawMotorFail
- ✓ Message 356 - Yaw Axis Motor Cmd
- ✓ Message 356 - Yaw Axis POST Motor



Indicates the Yaw servo motor failed the Power On Self Test (POST)

Corrective Action

- Check for failed Yaw servo motor
 - Short/open/high resistance between servo pins 3 and 4
- Check wiring between FGC and Yaw Servo
 - Shorts, opens, miswire
- Possible failed FGC



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 350 $\neq$ 0x00000:

✓ PB_ServoStatus

Look for other failed messages

✓ Message 351 $\neq$ 0x00000

✓ Message 357 $\neq$ 0x00000



BIT Results	
Message 350	0x00001
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00810
Message 365	0x00000
Message 360	0x00000

Message 350 Bits	
☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai

Select Board to Read CAN

☐ Board 1 ☒ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 351 $\neq$ 0x00000
 - ✓ Cont_DataMonFail



BIT Results	
Message 350	0x00001
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00810
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CallLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN
☐ Board 1 ☒ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 357 $\neq$ 0x00000
 - ✓ Pitch Trim Axis Motor Cmd
 - ✓ Pitch Trim Axis POST Motor



BIT Results			
Message 350	0x00001	Message 357 Bits	
Message 351	0x02000	☐ Yaw Trim Axis Insuffic	☐ Pitch Trim Axis Soleno
Message 352	0x00007	☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
Message 353	0xDF9F7	☐ Yaw Trim Axis Solenoid	☒ Pitch Trim Axis Motor
Message 354	0xFFBF6	☐ Yaw Trim Axis Motor Cm	☒ Pitch Trim Axis POST M
Message 355	0x00000	☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
Message 356	0x00000	☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
Message 357	0x00810	☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis Clutch
Message 365	0x00000	☐ Yaw Trim Axis Clutch E	☐
Message 360	0x00000	☐ Pitch Trim Axis Insuff	☐

Select Board to Read CAN

☐ Board 1 ☒ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 350 - PB_ServoStatus
- ✓ Message 357 – Pitch Trim Axis Motor
- ✓ Message 357 – Pitch Trim Axis POST M



Message 350 Bits

☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☒ PB_ServoStatus
☐ PB_Arinc6LoopFail	☐ PB_CalLevUserLimitsFai
	☐

Message 357 Bits

☐ Yaw Trim Axis Insuffic	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☒ Pitch Trim Axis Motor
☐ Yaw Trim Axis Motor Cm	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis Clutch
☐ Yaw Trim Axis Clutch E	☐
☐ Pitch Trim Axis Insuff	☐
	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

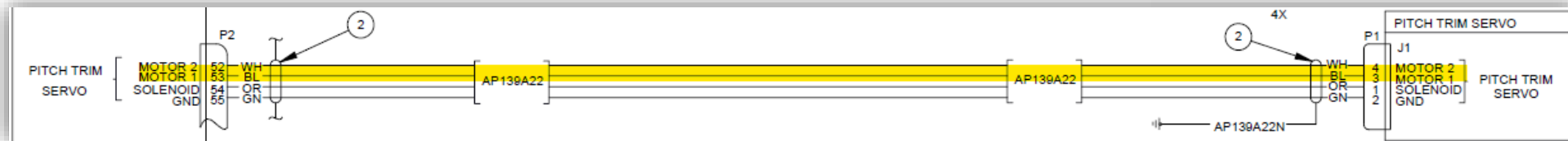
- ✓ Message 350 - PB_ServoStatus
- ✓ Message 357 – Pitch Trim Axis Motor
- ✓ Message 357 – Pitch Trim Axis POST M



Indicates the Pitch Trim servo motor failed the Power On Self Test (POST)

Corrective Action

- Check for failed Pitch Trim servo motor
 - Short/open/high resistance between servo pins 3 and 4
- Check wiring between FGC and Pitch Trim Servo
 - Shorts, opens, miswire
- Possible failed FGC



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 350 $\neq$ 0x00000:

✓ PB_CalLevUserLimitsFail

Look for other Failed Messages

✓ Message 351 $\neq$ 0x00000



BIT Results	
Message 350	0x00002
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 350 Bits	
☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CalLevUserCrcFail
☐ PB_Arinc5LoopFail	☐ PB_ServoStatus
☐ PB_Arinc6LoopFail	☒ PB_CalLevUserLimitsFai

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other Failed Messages

- ✓ Message 351 $\neq$ 0x00000
 - ✓ PB_CalLevUserLimitsFail
 - ✓ Cont_DataMonFail – (typically checked for AP FAIL conditions)



BIT Results	
Message 350	0x00002
Message 351	0x02000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☒ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 350 – PB_CallLevUserLimitsFail
- ✓ Message 351 – PB_CallLevUserLimitsFail



Message 350 Bits	
☐ PB_AppCrcFail	☐ PB_Arinc7LoopFail
☐ PB_WatchdogFail	☐ PB_Arinc8LoopFail
☐ PB_RamFail	☐ PB_EepromFail
☐ PB_Arinc1LoopFail	☐ PB_CalcSlopeFail
☐ PB_Arinc2LoopFail	☐ PB_EepromCrcFail
☐ PB_Arinc3LoopFail	☐ PB_CfgLimitsFail
☐ PB_Arinc4LoopFail	☐ PB_CallLevUserCrcFail
☐ PB_Arinc5LoopFail	☐ PB_ServoStatus
☐ PB_Arinc6LoopFail	☒ PB_CallLevUserLimitsFai
	☐

Message 351 Bits	
☒ PB_CallLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail
	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Failed Bits

- ✓ Message 350 – PB_CalLevUserLimitsFail
- ✓ Message 351 – PB_CalLevUserLimitsFail

- Indicates a bad IMU calibration on the selected board.

Corrective Action:

- Re-calibrate the IMU for the selected board using the ApCfgBuilder.
 - Reference the ICA document for proper calibration procedures



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 351 $\neq$ 0x00000:

- ✓ Cont_DataMonFail – Typically checked with many AP FAIL conditions
- ✓ Cont_PitMotorFail
- ✓ Cont_RollMotorFail
- ✓ Cont_YawMotorFail

Look for other Failed Messages

Message 352, 355 and 356 $\neq$ 0x00000



BIT Results	
Message 350	0x00000
Message 351	0x02070
Message 352	0x40007
Message 353	0xFF9F7
Message 354	0xFB8F7
Message 355	0x10100
Message 356	0x00100
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☒ Cont_PitMotorFail
☐ Cont_MisMonFail	☒ Cont_RollMotorFail
☒ Cont_DataMonFail	☒ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 352 – Cont_LowVoltageFail



BIT Results	
Message 350	0x00001
Message 351	0x02070
Message 352	0x40007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x15150
Message 356	0x00150
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 352 Bits	
☐ Cont_TrimFail	
☐ Cont_BezelFail	
☒ Cont_LowVoltageFail	
☐ Cont_AccelFail	
☐ Cont_5VRefFail	
☐	
☐	
☐	
☐	

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 355 – Multiple Pitch and Roll failures



BIT Results	
Message 350	0x00001
Message 351	0x02070
Message 352	0x40007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x15150
Message 356	0x00150
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 355 Bits	
☒ DM_PFD Configuration F	☐ Pitch Axis Clutch Enga
☐ SSM_354	☒ Roll Axis Insufficient
☒ Pitch Axis Insufficien	☐ Roll Axis Solenoid Ove
☐ Pitch Axis Solenoid Ov	☐ Roll Axis Solenoid Und
☐ Pitch Axis Solenoid Un	☐ Roll Axis Motor Cmd (V
☐ Pitch Axis Motor Cmd (	☒ Roll Axis POST Motor C
☒ Pitch Axis POST Motor	☐ Roll Axis POST Motor V
☐ Pitch Axis POST Motor	☐
☒ Pitch Axis POST Motor	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 356 – multiple Yaw failures



BIT Results			
Message 350	0x00001	Message 356 Bits	
Message 351	0x02070	☐ Roll Trim Axis Insuffi	☐ Yaw Axis Solenoid Over
Message 352	0x40007	☐ Roll Trim Axis Solenoi	☐ Yaw Axis Solenoid Unde
Message 353	0xDF9F7	☐ Roll Trim Axis Solenoi	☐ Yaw Axis Motor Cmd (Vo
Message 354	0xFFBF6	☐ Roll Trim Axis Motor C	☒ Yaw Axis POST Motor Cu
Message 355	0x15150	☐ Roll Trim Axis POST Mo	☐ Yaw Axis POST Motor Vo
Message 356	0x00150	☐ Roll Trim Axis POST Mo	☒ Yaw Axis POST Motor Vo
Message 357	0x00000	☐ Roll Trim Axis POST Mo	☐ Yaw Axis Clutch Engage
Message 365	0x00000	☐ Roll Trim Axis Clutch	☐
Message 360	0x00000	☒ Yaw Axis Insufficient	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 351 – Cont_PitMotorFail, Cont_RollMotorFail, Cont_YawMotorFail
- ✓ Message 352 – Cont_LowVoltageFail
- ✓ Message 355 – Pitch Axis Insufficient, Pitch Axis POST Motor, Pitch Axis Insufficient, Pitch Axis POST Motor, Roll Axis Insufficient, Roll Axis POST Motor
- ✓ Message 356 – Yaw Axis Insufficient, Yaw Axis POST Motor C, Yaw Axis Post Motor V



Message 351 Bits	Message 352 Bits	Message 355 Bits	Message 356 Bits
☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail	☒ DM_PFD Configuration F	☐ Roll Trim Axis Insuffi
☐ SSM_350	☐ Cont_AdcExt2Fail	☐ SSM_354	☐ Roll Trim Axis Solenoi
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail	☒ Pitch Axis Insufficien	☐ Roll Trim Axis Solenoi
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail	☐ Pitch Axis Solenoid Ov	☐ Roll Trim Axis Motor C
☐ Cont_SwVerFail	☐ Cont_AdcIntFail	☐ Pitch Axis Solenoid Un	☐ Roll Trim Axis POST Mo
☐ Cont_EnvMonFail	☒ Cont_PitMotorFail	☐ Pitch Axis Motor Cmd (	☐ Roll Trim Axis POST Mo
☐ Cont_MisMonFail	☒ Cont_RollMotorFail	☒ Pitch Axis POST Motor	☐ Roll Trim Axis POST Mo
☒ Cont_DataMonFail	☒ Cont_YawMotorFail	☐ Pitch Axis POST Motor	☐ Roll Trim Axis Clutch
☐ Cont_HardMonFail	☐ Cont_PitClutchFail	☒ Pitch Axis POST Motor	☒ Yaw Axis Insufficient
			☐ Yaw Axis Solenoid Over
			☐ Yaw Axis Solenoid Unde
			☐ Yaw Axis Motor Cmd (Vo
			☒ Yaw Axis POST Motor Cu
			☐ Yaw Axis POST Motor Vo
			☒ Yaw Axis POST Motor Vo
			☐ Yaw Axis Clutch Engage

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 351 – Cont_PitMotorFail, Cont_RollMotorFail, Cont_YawMotorFail
- ✓ Message 352 – Cont_LowVoltageFail
- ✓ Message 355 – Pitch Axis Insufficient, Pitch Axis POST Motor, Pitch Axis Insufficient, Pitch Axis POST Motor, Roll Axis Insufficient, Roll Axis POST Motor
- ✓ Message 356 – Yaw Axis Insufficient, Yaw Axis POST Motor C, Yaw Axis Post Motor V

- Indicates Board 1 no power, or out of range (LOW).

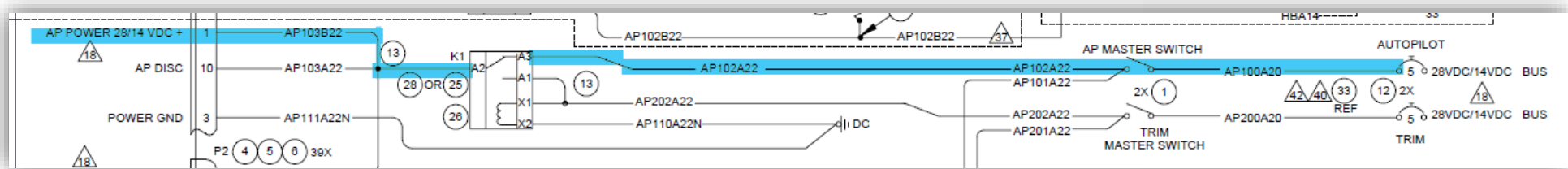
Corrective Action:

- Check power at P1-1.



Description	Data	Unit
Board 1 28V Power (Dbg37_0)	1630.998	value
Board 2 28V Power (Dbg37_1)	27980.19	value

Exit Refresh



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 351 $\neq$ 0x00000:

- ✓ Cont_SwVerFail
- ✓ Cont_DataMonFail

No other failed messages



BIT Results	
Message 350	0x00000
Message 351	0x42000
Message 352	0x00007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CallEvUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☒ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

Select Board to Read CAN
☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Failed Bits

- ✓ Cont_SwVerFail
- ✓ Cont_DataMonFail



Corrective Action:

- Check the CRC values using ApConfigBuilder.exe. The CRC values for Application, Configuration, and Install settings must match between board 1 and board 2.
 - Mismatched Application CRCs
 - re-load Application software to both boards
 - Mismatched Configuration CRCs
 - Re-load correct configuration file to both boards.
 - Mismatched Install CRCs
 - Re-do the install settings for both boards
- Remember to close TSOVista before opening ApCfgBuilder.exe

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 351 $\neq$ 0x00000:

✓ Cont_DataMonFail

Look for other failed messages

✓ Message 352 $\neq$ 0x00000



BIT Results	
Message 350	0x00000
Message 351	0x02000
Message 352	0x40007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 351 Bits	
☐ PB_CallLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Look for other failed messages

- ✓ Message 352 $\neq$ 0x00000
 - ✓ Cont_LowVoltageFail



BIT Results	
Message 350	0x00000
Message 351	0x02000
Message 352	0x40007
Message 353	0xDF9F7
Message 354	0xFFBF6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00000
Message 365	0x00000
Message 360	0x00000

Message 352 Bits	
☐ Cont_TrimFail	☐
☐ Cont_BezelFail	☐
☒ Cont_LowVoltageFail	☐
☐ Cont_AccelFail	☐
☐ Cont_5VRefFail	☐
☐	☐
☐	☐
☐	☐
☐	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Failed Bits

- ✓ Message 351 – Cont_DataMonFail
- ✓ Message 352 – Conf_LowVoltageFail



Message 351 Bits

☐ PB_CalLevUserLimitsFai	☐ Cont_AdcExt1Fail
☐ SSM_350	☐ Cont_AdcExt2Fail
☐ Cont_IntCanTxFail	☐ Cont_AdcExt3Fail
☐ Cont_MaxLoopFail	☐ Cont_AdcExt4Fail
☐ Cont_SwVerFail	☐ Cont_AdcIntFail
☐ Cont_EnvMonFail	☐ Cont_PitMotorFail
☐ Cont_MisMonFail	☐ Cont_RollMotorFail
☒ Cont_DataMonFail	☐ Cont_YawMotorFail
☐ Cont_HardMonFail	☐ Cont_PitClutchFail
	☐

Message 352 Bits

☐ Cont_TrimFail	☐
☐ Cont_BezelFail	☐
☒ Cont_LowVoltageFail	☐
☐ Cont_AccelFail	☐
☐ Cont_5VRefFail	☐
☐	☐
☐	☐
☐	☐
☐	☐

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

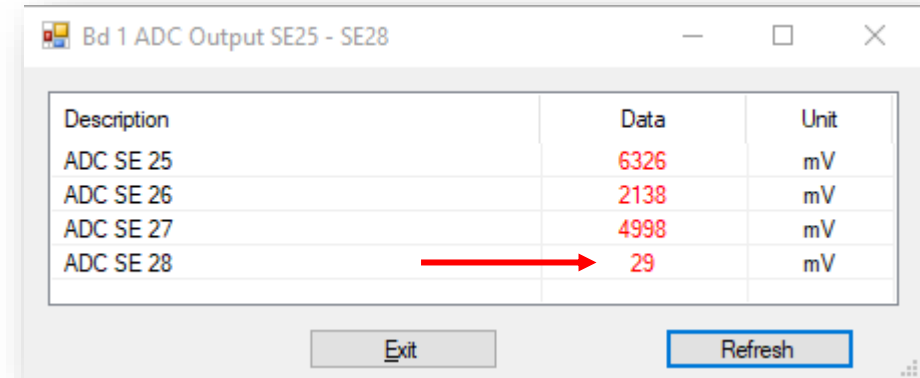
Failed Bits

- ✓ Message 351 – Cont_DataMonFail
- ✓ Message 352 – Conf_LowVoltageFail

- Indicates input voltage does not match 14/28 VDC strapping
- Refer to flag note 18

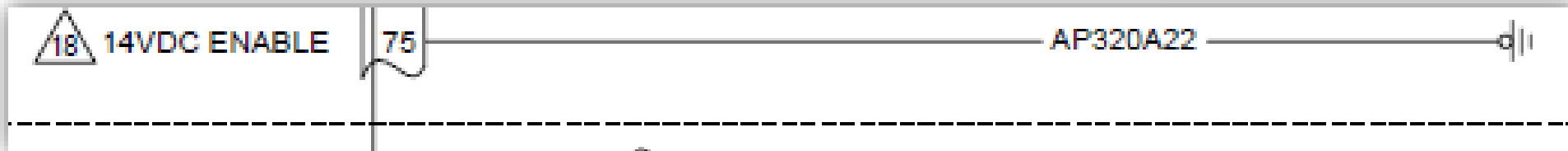
Corrective Action:

- Check P1-75 (ADC SE 28)
 - Grounded = 14 VDC operation
 - Open = 28 VDC operation

The image shows a software window titled 'Bd 1 ADC Output SE25 - SE28'. It contains a table with three columns: Description, Data, and Unit. The data for ADC SE 28 is highlighted with a red arrow pointing to the value '29'.

Description	Data	Unit
ADC SE 25	6326	mV
ADC SE 26	2138	mV
ADC SE 27	4998	mV
ADC SE 28	29	mV

Exit Refresh



AP COMPUTER POWER MAY BE 28VDC OR 14VDC DEPENDING ON THE AIRCRAFT'S GENERATED VOLTAGE. IF THE SYSTEM INSTALLATION REQUIRES 14 VDC OPERATION, CONNECTOR P1 PIN 75 MUST BE GROUNDED.

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Board 1 - Message 352 $\neq$ 0x00000:

✓ Cont_LowVoltageFail

Check for other failed messages

✓ Message 357 $\neq$ 0x00000



BIT Results	
Message 350	0x00000
Message 351	0x00000
Message 352	0x40007
Message 353	0x40000
Message 354	0x508F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00100
Message 365	0x00000
Message 360	0x00000

Message 352 Bits	
☐ Cont_TrimFail	☐
☐ Cont_BezelFail	☐
☒ Cont_LowVoltageFail	☐
☐ Cont_AccelFail	☐
☐ Cont_5VRefFail	☐
☐	☐
☐	☐
☐	☐
☐	☐

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 357 $\neq$ 0x00000:

✓ Pitch Trim Axis Insuff

No other failed messages



BIT Results

Message 350	0x00000
Message 351	0x00000
Message 352	0x40007
Message 353	0x40000
Message 354	0x508F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00100
Message 365	0x00000
Message 360	0x00000

Message 357 Bits

☐ Yaw Trim Axis Insuff	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Motor
☐ Yaw Trim Axis Motor Cm	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis Clutch
☐ Yaw Trim Axis Clutch E	
☒ Pitch Trim Axis Insuff	

Select Board to Read CAN

☒ Board 1 ☐ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Board 2 - Message 352 $\neq$ 0x00000:

✓ Cont_LowVoltageFail

Check for other failed messages

✓ Message 357 $\neq$ 0x00000



BIT Results	
Message 350	0x00000
Message 351	0x00000
Message 352	0x40007
Message 353	0x40000
Message 354	0x508F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00150
Message 365	0x00000
Message 360	0x00000

Message 352 Bits	
☐ Cont_TrimFail	☐
☐ Cont_BezelFail	☐
☒ Cont_LowVoltageFail	☐
☐ Cont_AccelFail	☐
☐ Cont_5VRefFail	☐
☐	☐
☐	☐
☐	☐
☐	☐

Select Board to Read CAN
☐ Board 1 ☒ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Message 357 $\neq$ 0x00000:

- ✓ Pitch Trim Axis Insuff
- ✓ Pitch Trim Axis POST M
- ✓ Pitch Trim Axis POST M



BIT Results	
Message 350	0x00000
Message 351	0x00000
Message 352	0x40007
Message 353	0x40000
Message 354	0x508F6
Message 355	0x00000
Message 356	0x00000
Message 357	0x00150
Message 365	0x00000
Message 360	0x00000

Message 357 Bits	
☐ Yaw Trim Axis Insuffic	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Motor
☐ Yaw Trim Axis Motor Cm	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis Clutch
☐ Yaw Trim Axis Clutch E	
☒ Pitch Trim Axis Insuff	

Select Board to Read CAN

☐ Board 1 ☒ Board 2

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Board 1 Failed Bits

- ✓ Message 352 – Conf_LowVoltageFail
- ✓ Message 357 – Pitch Trim Axis Insuff

Board 2 Failed Bits

- ✓ Message 352 – Conf_LowVoltageFail
- ✓ Message 357 – Pitch Trim Axis Insuff
- ✓ Message 357 – Pitch Trim Axis POST M
- ✓ Message 357 – Pitch Trim Axis POST M



Select Board to Read CAN

☒ Board 1 ☐ Board 2

BIT Results

Message 350	0x00000	
Message 351	0x00000	
Message 352	0x40007	
Message 353	0x40000	
Message 354	0x508F6	
Message 355	0x00000	
Message 356	0x00000	
Message 357	0x00100	
Message 365	0x00000	
Message 360	0x00000	

Message 352 Bits	
☐ Cont_TrimFail	
☐ Cont_BezelFail	
☒ Cont_LowVoltageFail	
☐ Cont_AccelFail	
☐ Cont_5VRefFail	

BIT Results

Message 350	0x00000	
Message 351	0x00000	
Message 352	0x40007	
Message 353	0x40000	
Message 354	0x508F6	
Message 355	0x00000	
Message 356	0x00000	
Message 357	0x00100	
Message 365	0x00000	
Message 360	0x00000	

Message 357 Bits	
☐ Yaw Trim Axis Insuffic	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Motor
☐ Yaw Trim Axis Motor Cm	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis Clutch E	☐ Pitch Trim Axis Clutch
☒ Pitch Trim Axis Insuff	

Select Board to Read CAN

☐ Board 1 ☒ Board 2

BIT Results

Message 350	0x00000	
Message 351	0x00000	
Message 352	0x40007	
Message 353	0x40000	
Message 354	0x508F6	
Message 355	0x00000	
Message 356	0x00000	
Message 357	0x00150	
Message 365	0x00000	
Message 360	0x00000	

Message 352 Bits	
☐ Cont_TrimFail	
☐ Cont_BezelFail	
☒ Cont_LowVoltageFail	
☐ Cont_AccelFail	
☐ Cont_5VRefFail	

BIT Results

Message 350	0x00000	
Message 351	0x00000	
Message 352	0x40007	
Message 353	0x40000	
Message 354	0x508F6	
Message 355	0x00000	
Message 356	0x00000	
Message 357	0x00150	
Message 365	0x00000	
Message 360	0x00000	

Message 357 Bits	
☐ Yaw Trim Axis Insuffic	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Soleno
☐ Yaw Trim Axis Solenoid	☐ Pitch Trim Axis Motor
☐ Yaw Trim Axis Motor Cm	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☒ Pitch Trim Axis POST M
☐ Yaw Trim Axis POST Mot	☐ Pitch Trim Axis POST M
☐ Yaw Trim Axis Clutch E	☐ Pitch Trim Axis Clutch
☒ Pitch Trim Axis Insuff	

TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista

Failed Bits

Board 1 Failed Bits

- ✓ Message 352 – Conf_LowVoltageFail
- ✓ Message 357 – Pitch Trim Axis Insuff

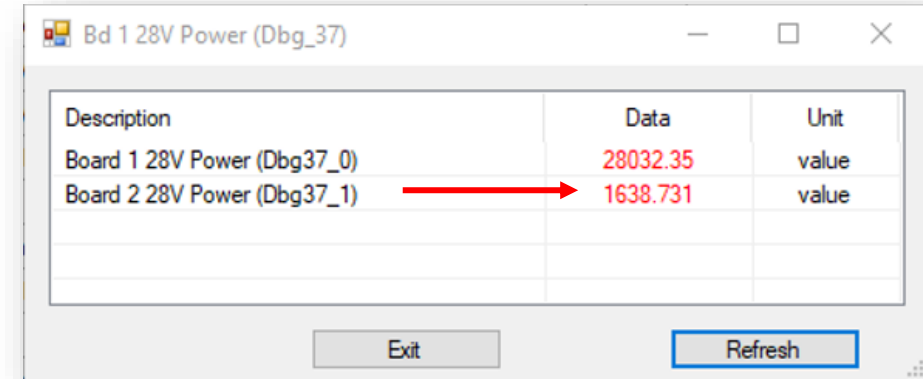
Board 2 Failed Bits

- ✓ Message 352 – Conf_LowVoltageFail
- ✓ Message 357 – Pitch Trim Axis Insuff
- ✓ Pitch Trim Axis POST M
- ✓ Pitch Trim Axis POST M

- Indicates Board 2 power insufficient (Trim servo failed POST)

Corrective Action:

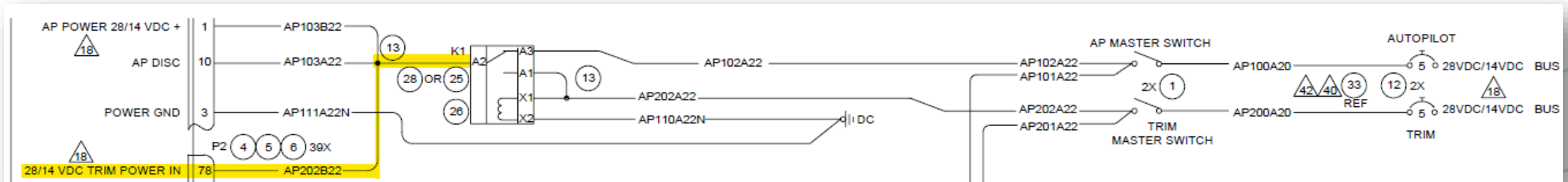
- Check P2-78



The screenshot shows a window titled "Bd 1 28V Power (Dbg_37)". It contains a table with three columns: Description, Data, and Unit. The data for Board 2 28V Power is highlighted in red, and a red arrow points to it from the text "Indicates Board 2 power insufficient (Trim servo failed POST)".

Description	Data	Unit
Board 1 28V Power (Dbg37_0)	28032.35	value
Board 2 28V Power (Dbg37_1)	1638.731	value

Buttons: Exit, Refresh



TROUBLESHOOTING AP FAIL (POST)

Troubleshooting an AP FAIL after POST message with TSOVista Hints

- If the FGC has the Yaw Damper option, be sure to check both Board 1 and Board 2 BIT Results.
- If you need help deciphering the Failed Message BIT's, email screenshots of the BIT Results in question, along with aircraft make/model and configuration information to:

US.AG.MW.support@moog.com



This is just a plug to remind me to inform everyone that we have released a training website.

Site: <https://MoogAvionics.litmos.com/self-signup/>

Password: Genesys Autopilot

S-TEC 3100 Digital Flight Control System



Questions & Answers